Project Title:

E-service platform for vehicle rentals

(BY SEMESTER –V OF III YEAR M.SC.(CA & IT) 2024-25)

Name of Company : Bhakti Tours And Travels(Bike For Rent)

Date of Submission : 07/09/2024

Submitted By:

Group ID : 23

Sr. No	Student Name	Roll No.
1	Mannan Agrawal	3049
2	Jainy Gandhi	3112
3	Diya Jain	3210

Submitted To:

K.S. School of Business Management

M.Sc. Computer Applications and Information Technology

Bhakti TOUR & TRAVELS

All type of Luxury Taxi Cars Available

Baji Raj Ka Kund, Inside Delhi Gate, Udaipur - 313 001 (Raj.) Mob. 99820-44434, 98292-44434

Ref. No Date

This is to certify that below mentioned students of K.S. School of Buisness Management (CA & IT) college worked as an intern as part of his/her Msc(CA & IT) course of Gujarat University .

The participants of the project trainee membership are given below.

Sr No.	Student Name	Student Roll-No.
1	Mannan Agrawal	3049
2	Jainy Gandhi	3112
3	Diya Jain	3210

Project Title : E-Service Platform for Vehicle Rentals

Hemant

Name: Hemant Agrawal
Designation: Owner
Date: 08/12/2023

Bhakti TOUR & TRAVELS

All type of Luxury Taxi Cars Available

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Ref. No Date

This is to certify that Ms. Diya Jain has successfully completed Analysis and Design of the project on 'E-Service platform for rental vehicle' under me. This work is original and has not been submitted elsewhere.

Hemant

Name: Hemant Agrawal
Designation: Owner
Date: 08/12/2023

Bhakti TOUR & TRAVELS

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Date

This is to certify that Mr. Mannan Agrawal successfully completed Analysis and Design of the project on 'E-Service platform for rental vehicle' under me. This work is original and has not been submitted elsewhere.

Hemant

Name: Hemant Agrawal
Designation: Owner
Date: 08/12/2023

Bhakti TOUR & TRAVELS

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This is to certify that Ms. Jainy Gandhi has successfully completed Analysis and Design of the project on 'E-Service platform for rental vehicle' under me. This work is original and has not been submitted elsewhere.

Hemant

Name: Hemant Agrawal
Designation: Owner
Date: 08/12/2023

Bhakti TOUR & TRAVELS

All type of Luxury Taxi Cars Available

Baiji Raj Ka Kund, Inside Delhi Gate, Udaipur - 313 001 (Raj.) Mob. 99820-44434, 98292-44434

Ref. No. Date

PROJECT COMPLETION CERTIFICATE

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Diya Kaivann Jain**, a student of M.Sc. (CA & IT), Semester VI, from K.S. School of Business Management & Information Technology, Gujarat University, has successfully completed their Project at '**Bhakti Tours And Travels**'.

The student has worked on the project titled:

"**E-service platform for vehicle rentals**"

The duration of the project was 9 Months.

This project represents the student's original work and has not been submitted elsewhere for any academic or professional purpose. During the project tenure, we found the student to be sincere, dedicated, technically proficient, and goal-oriented.

We appreciate the student's contribution to our organization and wish them continued success in all future endeavors.

Authorized Signatory
Name: Hemant Agrawal
Designation: Owner
Company Name: Bhakti tours and travels
Place: Udaipur
Date:29/04/2025

[Company Seal / Stamp]

Bhakti Tour & Travels

Proprietor

Bhakti TOUR & TRAVELS

All type of Luxury Taxi Cars Available

Baji Raj Ka Kund, Inside Delhi Gate, Udaipur - 313 001 (Raj.) Mob 99820-44434, 98292-44434

Ref No Date

PROJECT COMPLETION CERTIFICATE

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Jainy Chunang Gandhi**, a student of M.Sc. (CA & IT), Semester VI, from K.S. School of Business Management & Information Technology, Gujarat University, has successfully completed their Project at '**Bhakti Tours And Travels**'.

The student has worked on the project titled:

"**E-service platform for vehicle rentals**"

The duration of the project was 9 Months.

This project represents the student's original work and has not been submitted elsewhere for any academic or professional purpose. During the project tenure, we found the student to be sincere, dedicated, technically proficient, and goal-oriented.

We appreciate the student's contribution to our organization and wish them continued success in all future endeavors.

Authorized Signatory
Name: Hemant Agrawal
Designation: Owner
Company Name: Bhakti tours and travels
Place: Udaipur
Date:29/04/2025

[Company Seal / Stamp]

Bhakti Tour & Travels

Proprietor

Bhakti TOUR & TRAVELS

All type of Luxury Taxi Cars Available

Baji Raj Ka Kund Inside Delhi Gate Udaipur - 313 001 (Raj.) Mob 99820-44434, 98292-44434

Ref No Date

PROJECT COMPLETION CERTIFICATE

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This is to certify that **Mannan ManishKumar Agrawal**, a student of M.Sc. (CA & IT), Semester VI, from K.S. School of Business Management & Information Technology, Gujarat University, has successfully completed their Project at '**Bhakti Tours And Travels**'.

The student has worked on the project titled:

"**E-service platform for vehicle rentals**"

The duration of the project was 9 Months.

This project represents the student's original work and has not been submitted elsewhere for any academic or professional purpose. During the project tenure, we found the student to be sincere, dedicated, technically proficient, and goal-oriented.

We appreciate the student's contribution to our organization and wish them continued success in all future endeavors.

Authorized Signatory
Name: Hemant Agrawal
Designation: Owner
Company Name: Bhakti tours and travels
Place: Udaipur
Date:29/04/2025

[Company Seal / Stamp]

Bhakti Tour & Travels

Proprietor

ACKNOWLEGEMENT

We would like to express our heartfelt thanks to our mentor, **Mr. Pritesh Vyas** Sir, for his constant support and guidance throughout this project. His valuable advice and encouragement have played a crucial role in helping us move forward and complete this work successfully.

We would also like to thank **Bhakti Tours and Travels** for their generous help and cooperation. They provided us with useful information and resources that greatly improved the quality of our project.

We also express our sincere thanks to all our friends & beneficial who helped us throughout this project.

Thanking You,

3210 - Diya Jain

3112 - Jainy Gandhi

3049 - Mannan Agrawal

INDEX

Chapter Name	Pg. No.
1.Introduction…….....	11
1.1 Organization Profile……	11
1.2 System Details…….	11
1.2.1 Existing System…….	11
1.2.2 Proposed System…….....	11
1.3 Scope Of System…….	12
1.4 Objectives…….....	13
2.Proposed System Requirement Gathering……	14
2.1 Stakeholder Of System…….....	14
2.2 Requirement Gathering Technique Used……	15
2.3 Consolidated List Of Requirement……..	18
2.4 Project Definition…….....	18
3.System Management And Planning	20
3.1 Feasibility Study…….	20
3.1.1 Technical Study……	20
3.1.2 Economical Study…….	20
3.1.3 Operational Study……..	21
3.2 Hardware-Software Requirements……..	22
3.3 System Planning……..	23
3.3.1 Work Breakdown Structure……..	23
3.3.2 Gantt Chart……	24
3.4 Process Model…….	25
4.System Analysis And Design	27
4.1 UML(Unified Modeling Language)……	27
4.1.1 Use Case Diagram……	27
4.1.2 Activity Diagram……..	31
4.1.3 Class Diagram……..	35
4.1.4 Sequence Diargam……...	37
4.2 System Flow Diagram…….	40

Chapter Name	Pg. No.

4.3 Data Dictionary……. 42
4.4 User Interface……. 47
4.5 System Navigation…… 50
5.Input/Output Design 51
6.Testing 62
7.Summary 65
7.1 Assumptions…….. 65
7.2 Limitations…… 66
7.3 Future Scope……. 67
7.4 Conclusion…… 68
Bibliography …… 69
Evaluation Form…… 70

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Chapter-1
Introduction

1.1 Organization Profile

Company Name	Bhakti tours and travels
Owner Name	Hemant Agrawal
Address	11,Shivaji Nagar , Inside Udaipole, Udaipur-313 001, Rajasthan.
E-Mail	udaipurtaxihire@gmail.com
Phone No.	9829244434.

Organization Overview :

Bhakti Tours and Travel is a company in Udaipur that offers rental vehicles for tourists.
It provides a variety of options, including scooters and bikes, to help visitors explore the city comfortably. The goal of them is to make traveling around Udaipur easy and enjoyable by offering reliable vehicles and friendly service.

1.2 System Details

1.2.1 Existing System

- Existing system operates entirely **offline**, requiring in-person or phone interactions.
- Employees **manually provide vehicle information** and availability, which is **time-consuming**.
- Booking process requires **customers to visit or call**, with manual checks for availability.
- Customer convenience is **limited to business hours**, offering **less flexibility**.
- Resource management is manual, leading to potential **errors and inefficiencies**.

1.2.2 Proposed System

- Proposed e-service web portal will be **accessible from everywhere**.
- Website will display all vehicle details and availability, **reducing the need for employee intervention.**
- Customers can book vehicles online, streamlining the process and **reducing manual checks**.
- Portal offers flexibility, allowing customers **to access services 24/7**.
- Automated management of bookings and availability improves **accuracy and efficiency, saving time**.

1.3 Scope Of System

- Online Platform : Build a 24/7 accessible website.
- Information Sharing : Display vehicle details and rates online.
- Online Booking : Enable real-time vehicle checks and bookings.
- Automated Management : Automatically update availability and bookings.
- Convenience Features : Offer easy navigation, secure payments, and instant confirmations.
- Backend Integration : Connect online and offline systems.
- Feedback System : Allow customer reviews to improve service.
- Price Management : Adjust prices based on demand and season(by the admin).
- GPS Tracking : Add GPS to track vehicle locations and provide real-time updates(by the admin).

1.4 Objectives

- We aim to create a simple and **user-friendly website** that is **accessible 24/7**, allowing customers to access rental services **anytime and from anywhere**.
- The website will **display all necessary details** about vehicles, rental rates, and terms, reducing the need for manual communication and making information easily accessible.
- Customers will be able to **check vehicle availability**, select their dates, and book vehicles directly through the website.
- We will implement a system that **automatically updates vehicle availability and manages bookings**, helping to avoid mistakes and **prevent double bookings**.
- The website will include features that enhance the customer experience, such as **easy navigation**, **secure payment options**, and **instant booking confirmations**.
- The new online platform will be integrated with existing offline operations to ensure **consistent and well-managed data.**

Chapter-2
Proposed System Requirement Gathering

2.1 Stakeholders Of The System

1. Admin Requirements :

- **Dashboard :** Real-time overview of bookings, payments, and customer queries.
- **User/Vehicle Management :** Manage user accounts, vehicle listings, and maintenance schedules.
- **Booking/Transaction Management :** Monitor and manage bookings, payments, cancellations, and refunds.
- **Content Management :** Update website content, pricing, and promotions.
- **Customer Support :** Integrated tools for handling queries and feedback.
- **Integrations :** Connect with GPS, payment gateways.

2. User Requirements(Registered Customers) :

- **Account Management :** Create/manage accounts, login via email.
- **Search/Booking :** Search, filter, and book vehicles with ease.
- **Payments :** Secure online payment options and view transaction history.
- **Booking Management :** Modify or cancel bookings
- **Reviews/Feedback :** Leave ratings and feedback.

3. Visitor Requirements (Unregistered Users) :

- **Information Access :** View vehicle options, pricing, and services without registering.
- **Search/Inquiry :** Basic search and booking inquiry options.
- **Promotions :** Access to current discounts and offers.
- **Registration Prompt :** Encourage sign-ups for better access and discounts.
- **Security/Cancellation :** Clear policy and terms of service.
- **Reviews/Feedback :** Leave ratings and feedback.

2.2 Requirement Gathering Technique Used

1. ADMIN :

- **Q1:** Do you currently face challenges in managing vehicle bookings and user accounts? **Ans:** Yes
- **Q2:** Is customer interaction primarily managed through manual processes? **Ans:** Yes
- **Q3:** Do you need a real-time dashboard for monitoring bookings and payments? **Ans:** Yes
- **Q4:** Is it important for you to generate regular reports on revenue and customer feedback? **Ans:** Yes
- **Q5:** Do you need the ability to manage user accounts directly from the admin panel? **Ans:** Yes
- **Q6:** Do you require a system to handle booking cancellations and refunds? **Ans:** Yes
- **Q7:** Do you regularly update vehicle details and pricing on the platform? **Ans:** Yes
- **Q8:** Do you currently track customer support issues manually? **Ans:** Yes
- **Q9:** Do you currently use any third-party services like GPStracking or payment gateways? **Ans:** Yes
- **Q10:** Are you planning to expand the platform's features in the future? **Ans:** Yes

2. USER (REGISTERED CUSTOMERS):

- **Q1:** Would you create an account if it provided benefits like faster booking or exclusive offers?

  Ans:Yes

- **Q2:** Is a user-friendly interface important to you when using an online platform?

  Ans:Yes

- **Q3:** Is having the ability to search and filter vehicles by type, price, or availability important to you?

  Ans:Yes

- **Q4:** Would you use a platform that allows you to book vehicles online?

  Ans:Yes

- **Q5:** Would you prefer to have multiple payment options (e.g., credit/debit cards, UPI, wallets) available?

  Ans:Yes

- **Q6:** Is receiving a detailed receipt after payment important to you?

  Ans:Yes

- **Q7:** Would you want the option to modify or cancel your bookings after they are made?

  Ans:Yes

- **Q8:** Would you like the option to track your rented vehicle's location using GPS?

  Ans:Yes

- **Q9:** Would you expect to have access to customer support for booking-related issues?

  Ans:Yes

- **Q10:** Would you be likely to leave a review or provide feedback after using the service?

  Ans:Yes

3. VISITOR (UNREGISTERED CUSTOMERS) :

- **Q1:** Would you like to view vehicle options and pricing information without needing to register?

  Ans: Yes
- **Q2:** Is it important for you to see detailed descriptions and photos of vehicles before making a decision?

  Ans: Yes
- **Q3:** Would you use a search function to find specific vehicles or services on the platform?

  Ans: Yes
- **Q4:** Would you be interested in seeing promotional offers or discounts available on the platform?

  Ans: Yes
- **Q5:** Is it important for you to be notified of special deals or limited-time offers?

  Ans: Yes
- **Q6:** Is a simplified registration process important to you?

  Ans: Yes
- **Q7:** Would you find a user-friendly website layout essential for browsing vehicle options and services?

  Ans: Yes
- **Q8:** Is it important for you to have a fast-loading website with easy navigation?

  Ans: Yes
- **Q9:** Would you be likely to return to the platform for future vehicle rentals if your initial experience was positive?

  Ans: Yes
- **Q10:** Does Secure Payment Methods make you register into the page?

  Ans: Yes

2.3 Consolidated List Of Requirements

- Real-time monitoring, user/vehicle management, and maintenance tracking.
- Search, filter, and book vehicles online, with accessible details for visitors.
- Support for multiple payment options, transaction history, and refund handling.
- Regular updates of vehicle info and promotional offers.
- Integrated support system, FAQs and inquiry options.
- GPS tracking and real-time booking updates.
- Fast, user-friendly, secure platform with easy registration.
- Option to leave reviews and receive follow-up.
- Integration with GPS and payment services, and scalable for future growth.

2.4 Project Definition

The project aims to modernize **Bhakti Tours and Travels** by developing a comprehensive online platform for **vehicle rentals**. This platform will allow users to search for, book, and pay for vehicle rentals easily and securely. The design will focus on creating intuitive and user-friendly interfaces for users, visitors, and administrators, ensuring that the platform is accessible and efficient. **Key features will include real-time vehicle search and booking, multiple payment options, and GPS tracking to enhance the user experience.**

Administrators will benefit from a powerful dashboard that enables them to manage user accounts, vehicle listings, and bookings efficiently. The dashboard will also provide tools for

monitoring real-time activity and managing customer support inquiries. Additionally, the platform will include integrated customer support features to address user concerns promptly and improve overall service quality.

Furthermore, the system will be designed with scalability in mind, allowing for future growth and the addition of new features as Bhakti Tours and Travels expands its services. Overall, this project will provide a **modern and efficient solution** for the company's vehicle rental operations.

Chapter-3
System Management And Planning

3.1 Feasibility Study

3.1.1 Technical Study

Components	Technology	Description
Frontend	React	This framework allows for building interactive user interfaces and provides flexibility and ease of use, ensuring a smooth user experience.
Backend	Node.js and Express	Offers scalability and efficient handling of server requests. This ensures that the system can manage increased demand as it grows.
Database	MongoDB	A NoSQL database that is flexible and can efficiently handle various types of data. It allows easy scaling and is a good fit for managing vehicle records and bookings

3.1.2 Economical Study

The main development costs are reduced due to the open-source nature of the MERN stack, meaning no licensing fees are required for MongoDB, React, Node.js, or Express. The primary

expenses will include server hosting, which can be minimized using cloud services like AWS or Heroku. Long-term maintenance costs will be limited to occasional updates and minor operational fees for server uptime. Overall, the cost of moving from a manual booking system to an online one is minimal compared to the increase in efficiency and potential revenue.

3.1.3 Operational Study

- The new system will greatly improve user experience by offering 24/7 access to vehicle bookings, reducing the need for manual intervention.
- Customers will be able to search for available vehicles, book, and make payments directly online, which simplifies the process.
- Staff will also benefit from a user-friendly admin panel to manage bookings and vehicle details.
- The new system will integrate seamlessly with existing operations, making it easy to adopt and improving overall business efficiency.

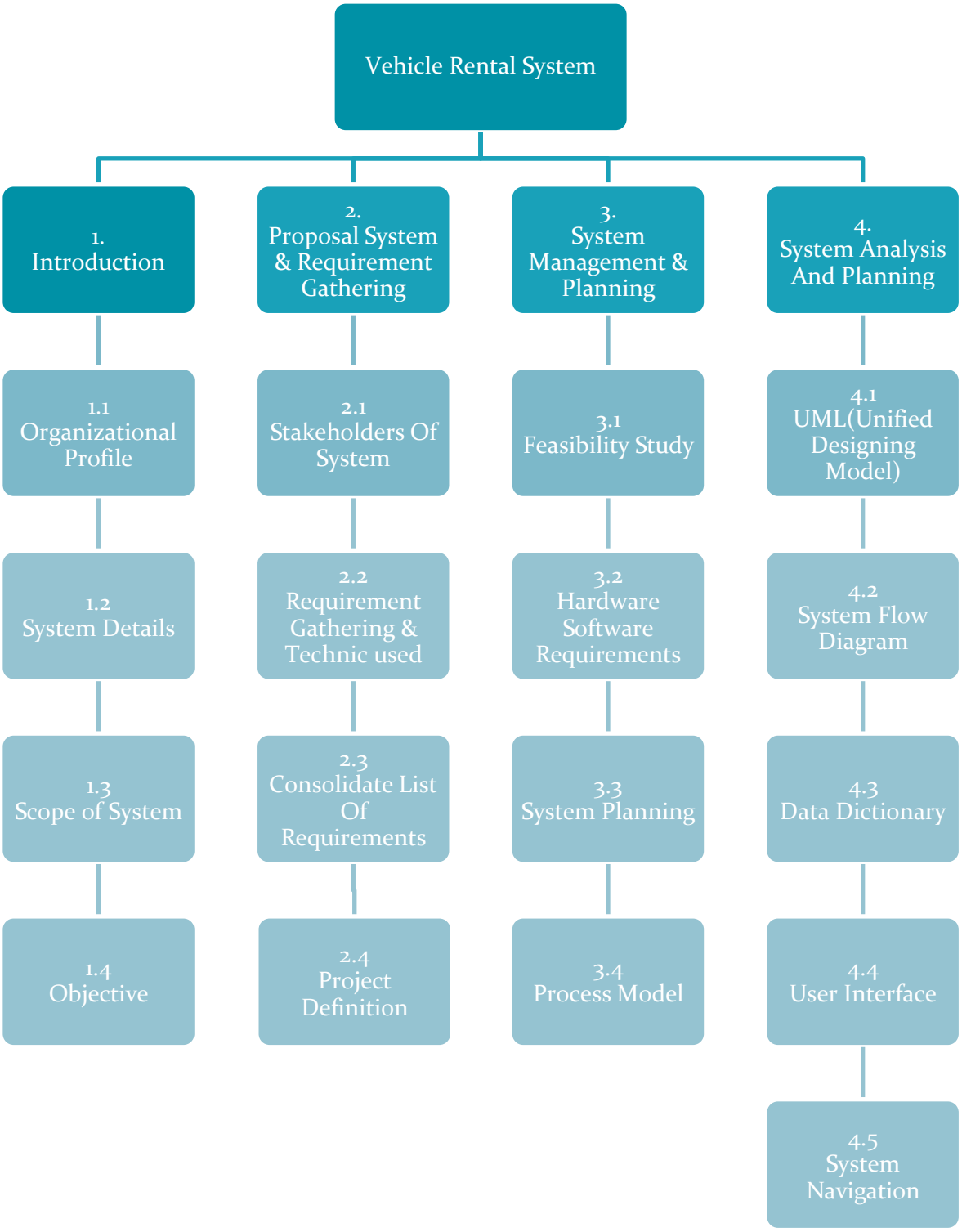
3.2 Hardware-Software Requirements

Client-Side Requirements	
Hardware Requirements	
Desktop/Laptop	Minimum 4 GB RAM
Processor	Minimum Core i3 or above
Storage	Minimum 20 GB or higher
Software Requirements	
Operating System	Windows 7 or higher
Browser	Chrome version 73.0.3683.86 or higher

Server-Side Requirements	
Hardware Requirements	
Processor	Minimum Core i3 or above
Ram	Minimum 2 GB
Storage	Minimum 500 MB
Software Requirements	
Backend	Node.js
Database	MongoDB
Frontend	React
Backend	Express.Js

3.3 System Planning

3.3.1 Work Break Down Structure



3.3.2 Gantt Chart

Task Processor	August	September	October	November	December	January	February	March
1. Introduction								
➢ Organisation Profile								
➢ System Details								
➢ Scope Of System								
➢ Objectives								
2. Proposed System Requirement Gathering								
➢ Stakeholder of System								
➢ Requirement Gathering Technique Used								
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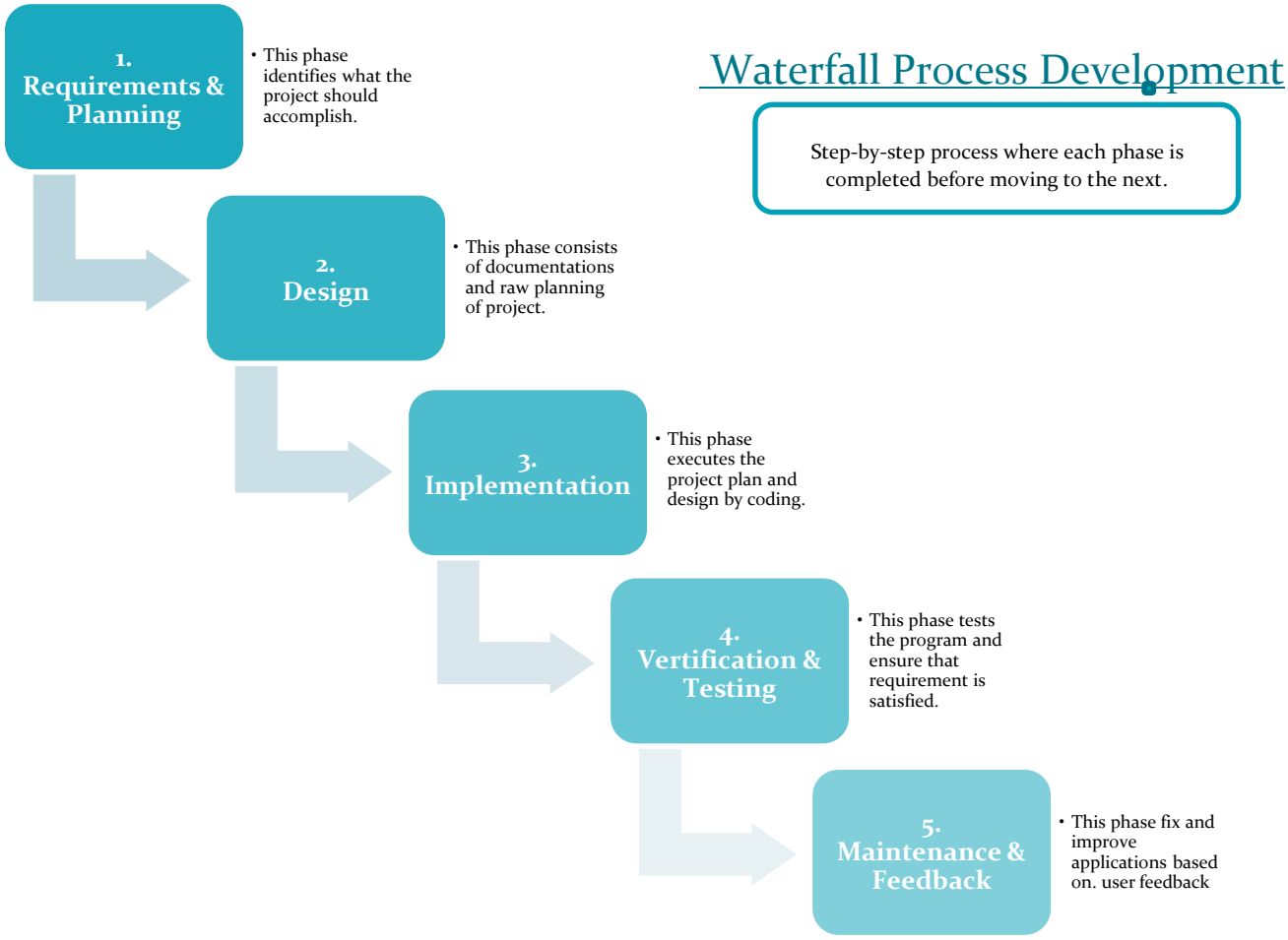
➢ Process Model								
4. System Analysis And Planning								
➢ UML Diagram								
➢ System Flow Diagram								
➢ User Interface								
➢ System Navigation								
5.I/O Design								
6.Testing								
7.Summary								

3.4 Process Model



Why This Model :

We chose the Waterfall model for this project because it allows us to work step-by-step. Since the project requires clear features like vehicle booking, payments, and GPS tracking, we can plan everything at the start. Then, we move to the next phase only after completing the previous one. This approach helps us ensure that the platform is developed properly without missing any key details. It's a simple and effective method, ideal for projects where all the requirements are known in advance, like this rental platform for Bhakti Tours and Travels.

Stages :

1. **Requirements and Planning**
   In the first step, we figure out what the vehicle rental website needs. We list all the features, like searching for cars, booking them, making payments, and using GPS. This helps everyone know what we want to create.

2. **Design**
   Next, we design how the website will look. We draw simple pictures of each page and plan how users will move around the site. We also decide how the backend will work, like how we'll store information. This design acts as our guide for building the site.

3. **Implementation**
   In this step, we actually build the website. We write code to make the front end and the back end . This is where everything starts coming together.

4. **Verification and Testing**
   After building, we test the website to make sure everything works. We check if users can easily book vehicles and if payments go through. This step is important to catch any mistakes before people start using it.

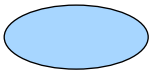
5. **Maintenance and Feedback**
   Once the website is live, we keep watching it for any issues. We help users if they have questions and listen to their feedback to make improvements. This way, we can keep the website running smoothly and make it better over time.

Chapter-4
System Analysis And Design

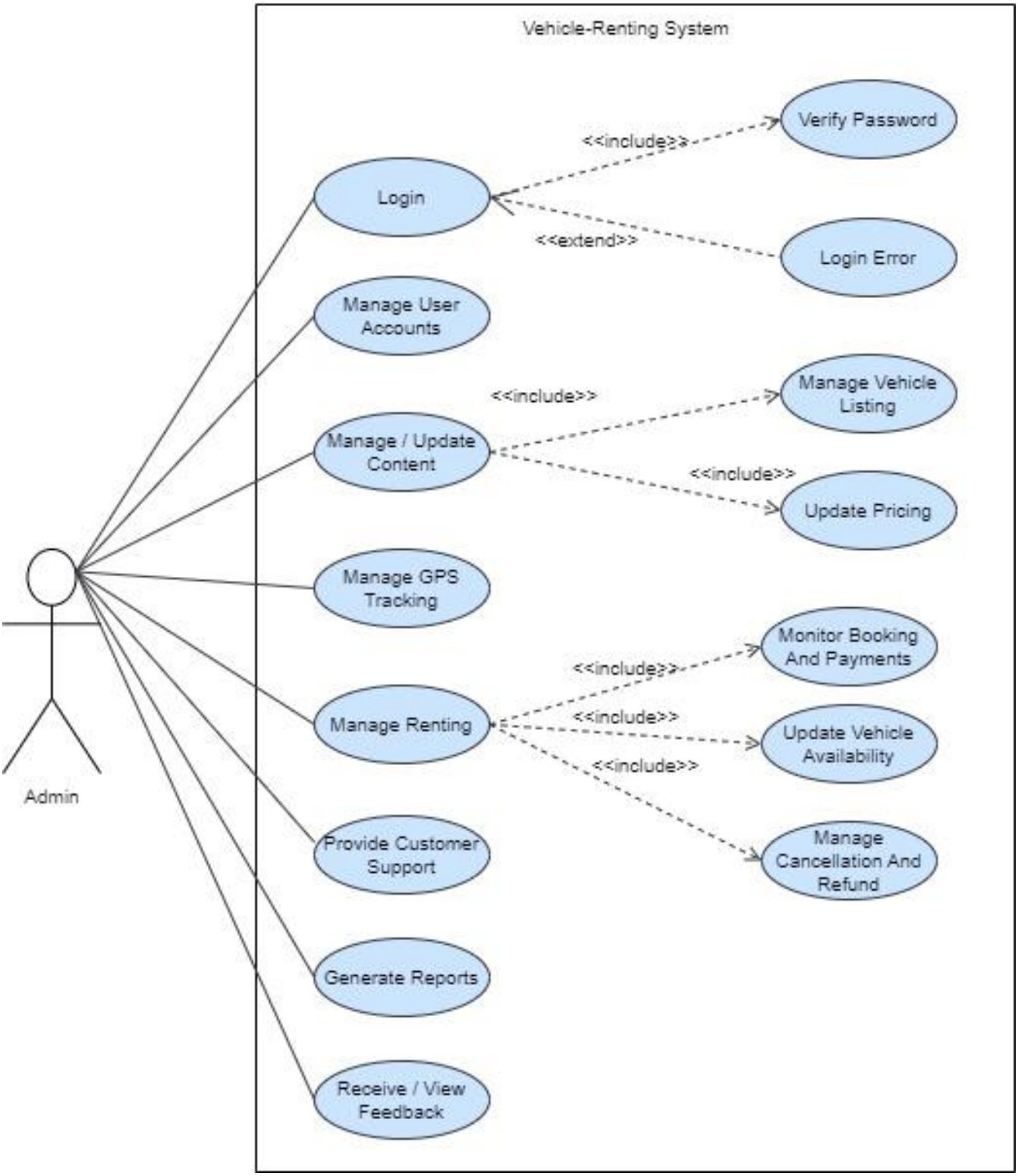
4.1 UML(Unified Modelling Language)

4.1.1 Use Case Diagram

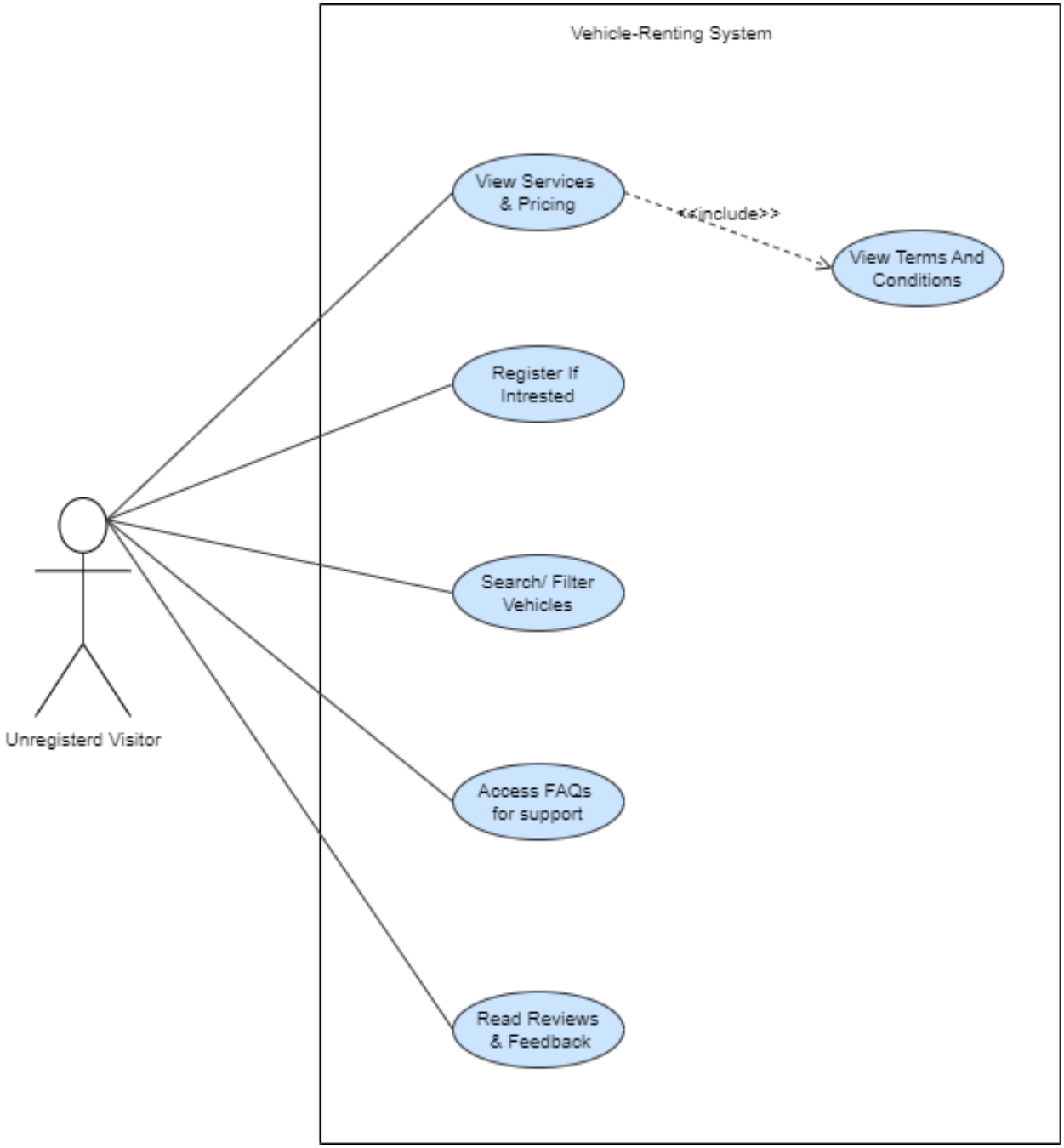
An **use case** diagram visually represents the interactions between users (actors) and a system's functions (use cases). It shows who can perform which actions and highlights relationships among use cases, helping to define system requirements and user roles.

Symbol	Description
	System Boundary
	Actor
	Use Cases
————————	Association
---------->	Include
<----------	Extend

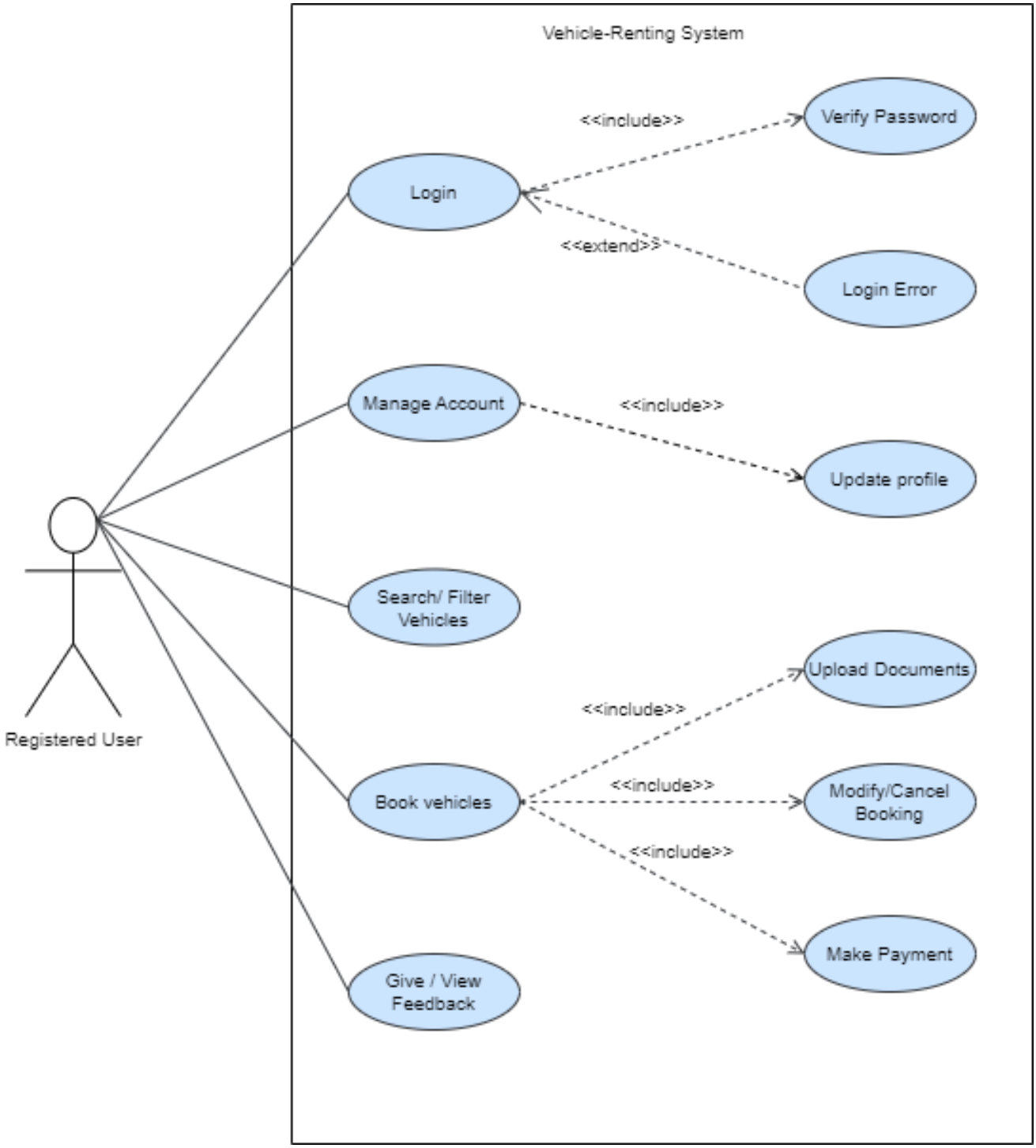
Admin:


Vehicle-Renting System
Verify Password
<<include>>
Login
<<extend>>
Login Error
Manage User Accounts
Manage Vehicle Listing
<<include>>
Manage / Update Content
<<include>>
Update Pricing
Manage GPS Tracking
Monitor Booking And Payments
<<include>>
Manage Renting
<<include>>
Update Vehicle Availability
<<include>>
Manage Cancellation And Refund
Provide Customer Support
Generate Reports
Receive / View Feedback
Admin

Visitor :

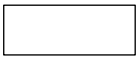
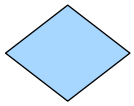

Vehicle-Renting System
View Services & Pricing
<<include>>
View Terms And Conditions
Register If Intrested
Search/ Filter Vehicles
Access FAQs for support
Read Reviews & Feedback
Unregisterd Visitor

Registered User :

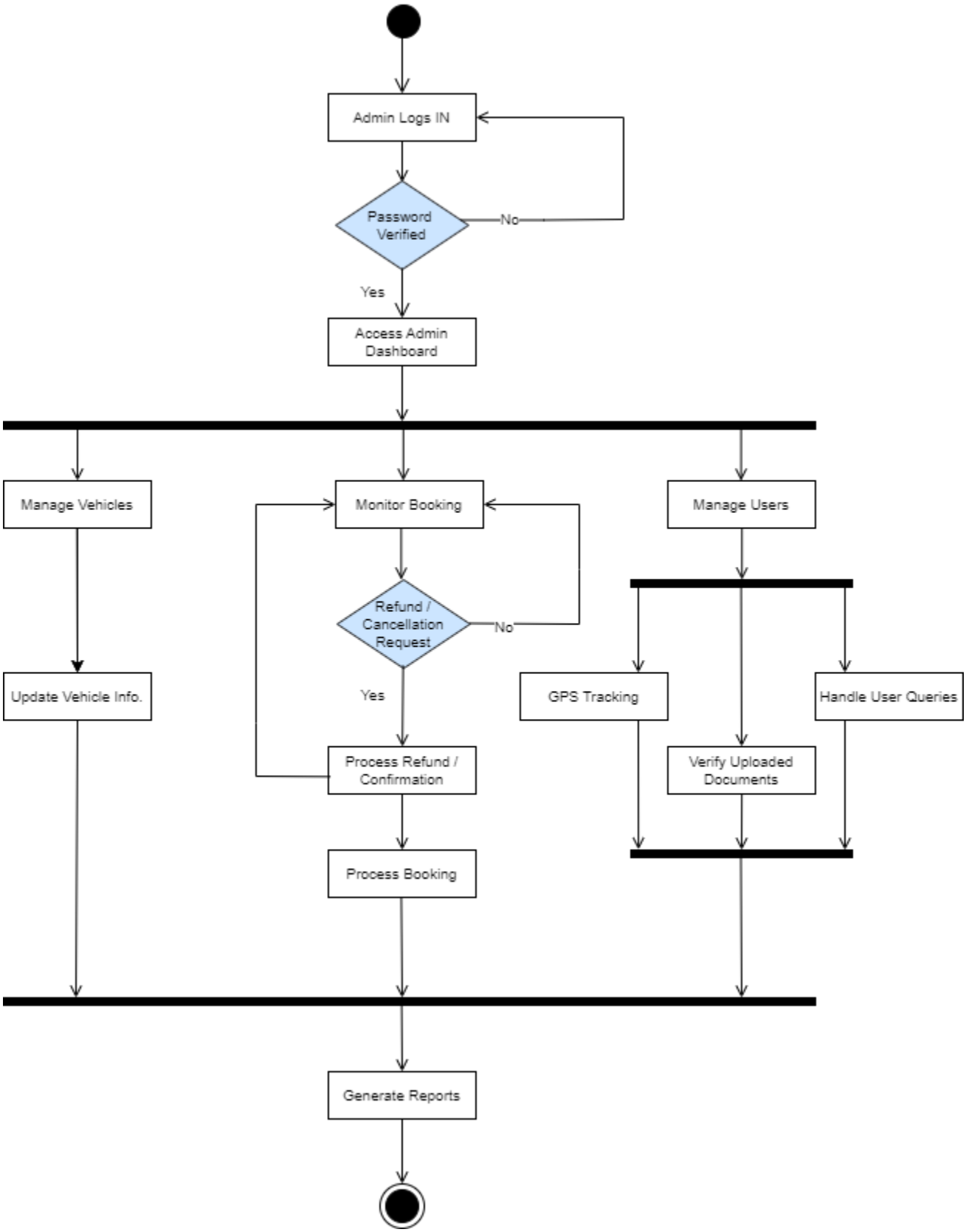


4.1.2 Activity Diagram

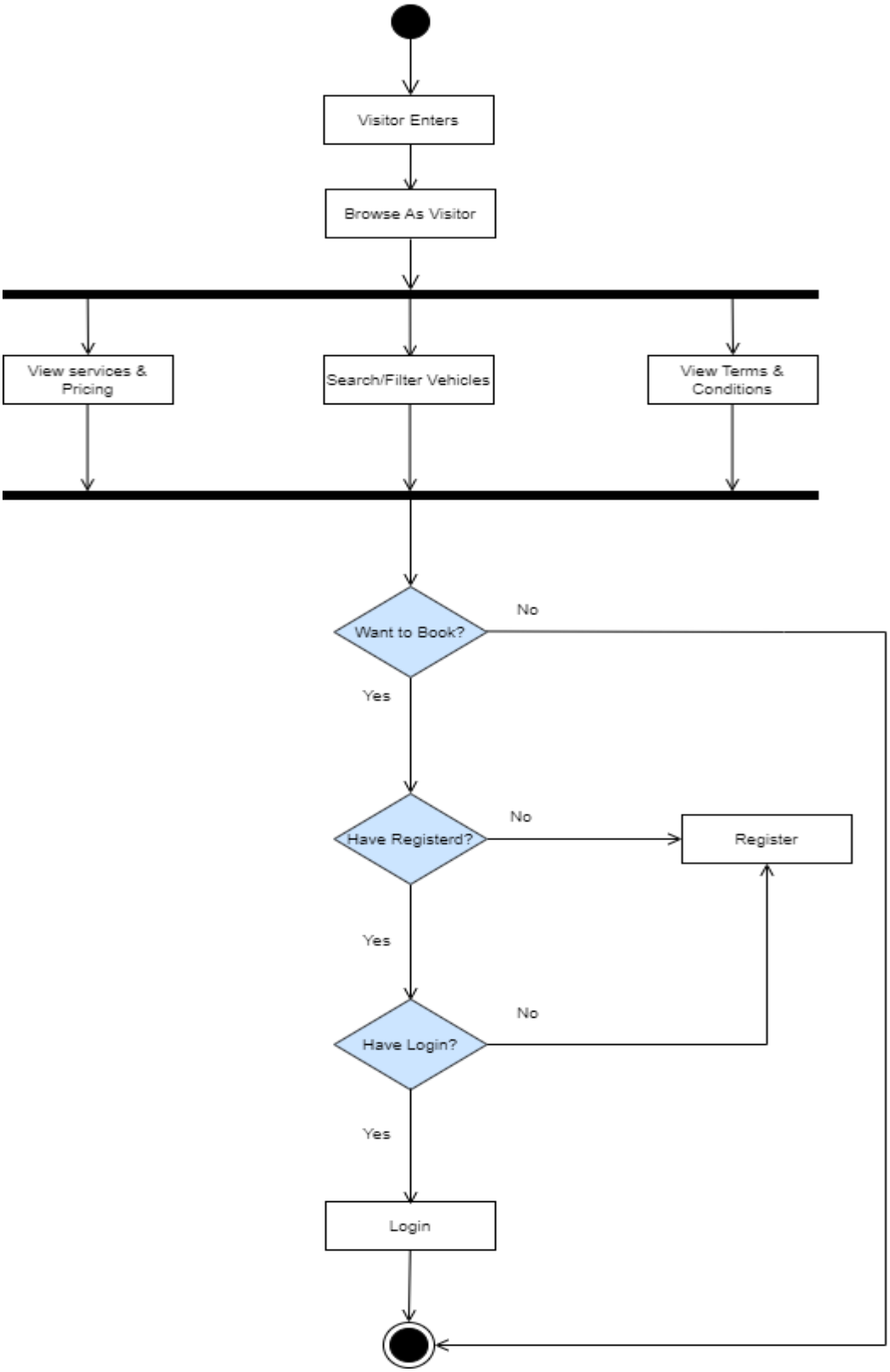
An **activity diagram** is a flowchart-like UML diagram that models the sequence of activities or tasks in a process. It captures the flow of control, decisions, and concurrent actions, helping to illustrate the detailed workflow within a system.

Symbol	Description
	Start Node
	Activity Node
	Decision Node
	Fork / Join Node
	Control Flow
	End Node

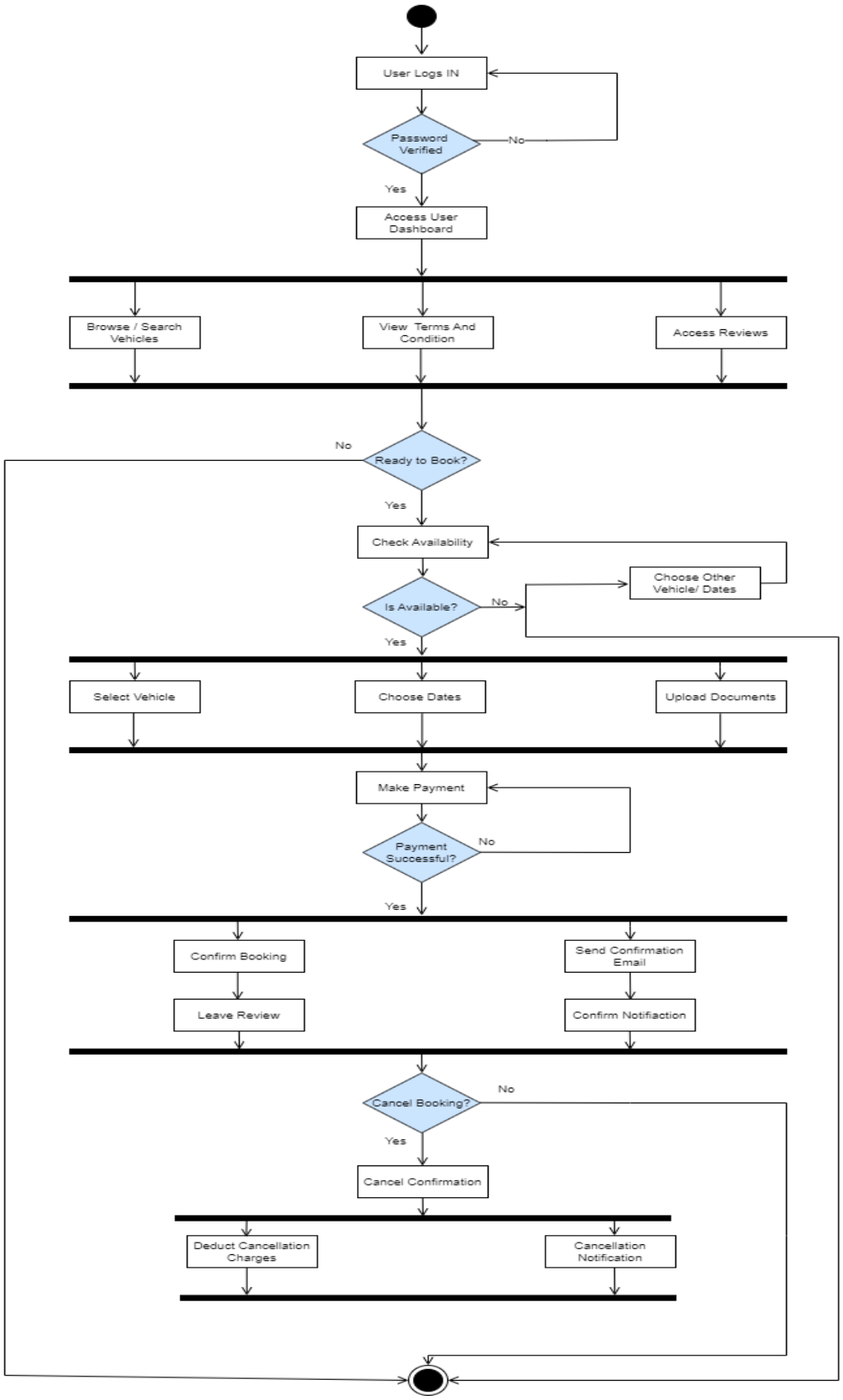
Admin:


Admin Logs IN
Password Verified
No
Yes
Access Admin Dashboard
Manage Vehicles
Monitor Booking
Manage Users
Refund / Cancellation Request
No
Yes
Update Vehicle Info.
GPS Tracking
Handle User Queries
Process Refund / Confirmation
Verify Uploaded Documents
Process Booking
Generate Reports

Visitor :

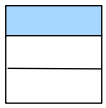
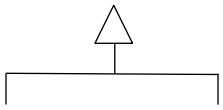
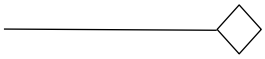
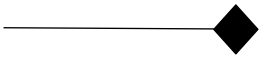

Visitor Enters
Browse As Visitor
View services & Pricing
Search/Filter Vehicles
View Terms & Conditions
Want to Book?
No
Yes
Have Registerd?
No
Register
Yes
Have Login?
No
Yes
Login

Registered User :

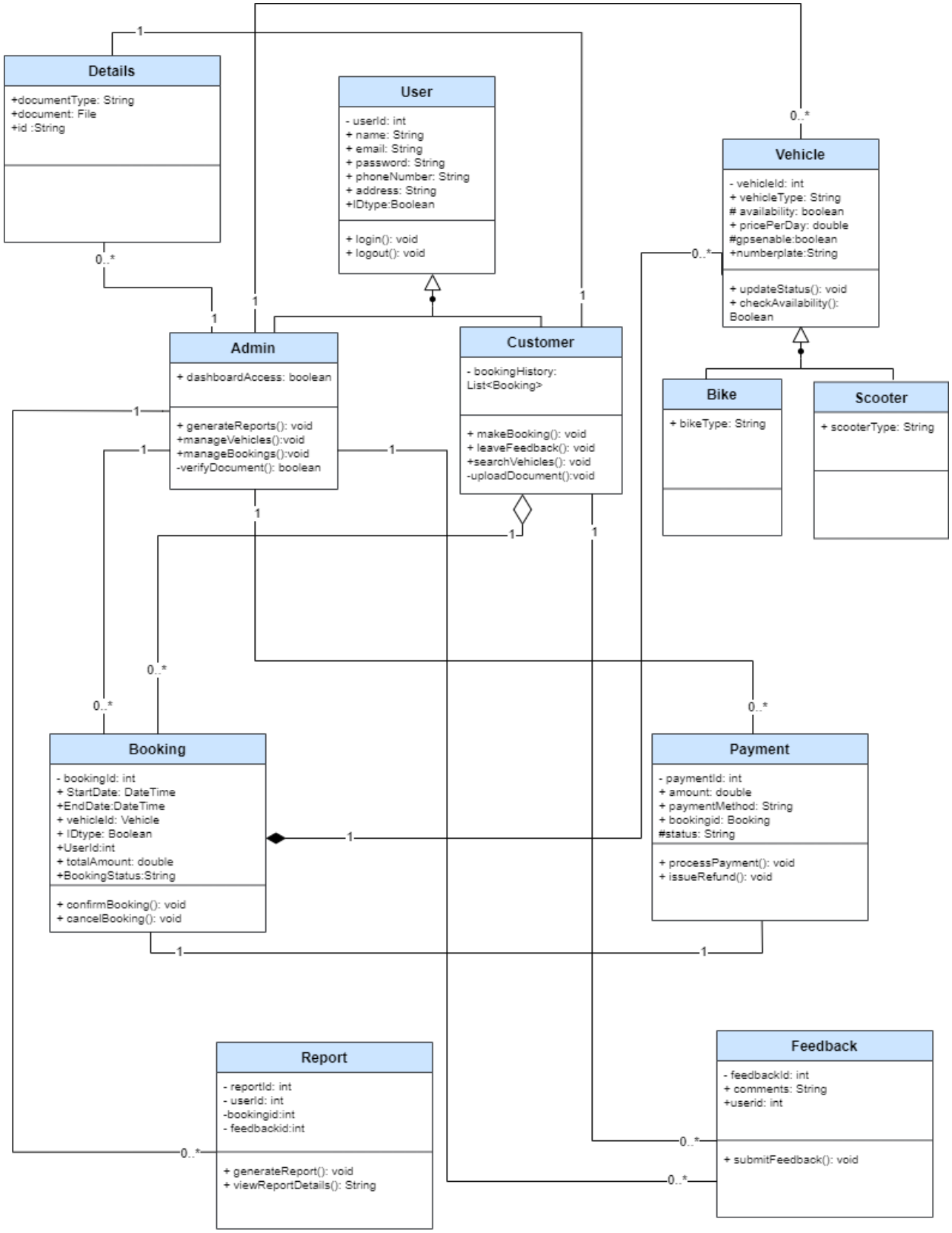


4.1.3 Class Diagram

A **class diagram** is a UML diagram that models the structure of a system by showing its classes, attributes, methods, and relationships. It defines how different classes interact and provides a blueprint of the system's objects and data structure.

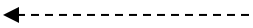
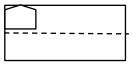
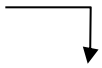
Symbol	Description
	Class
	Association
	Generalization
	Aggregation
	Composition
0..*	0 to many multiplicity
1..*	1 to many multiplicity
0..1	0 to 1 multiplicity

System :

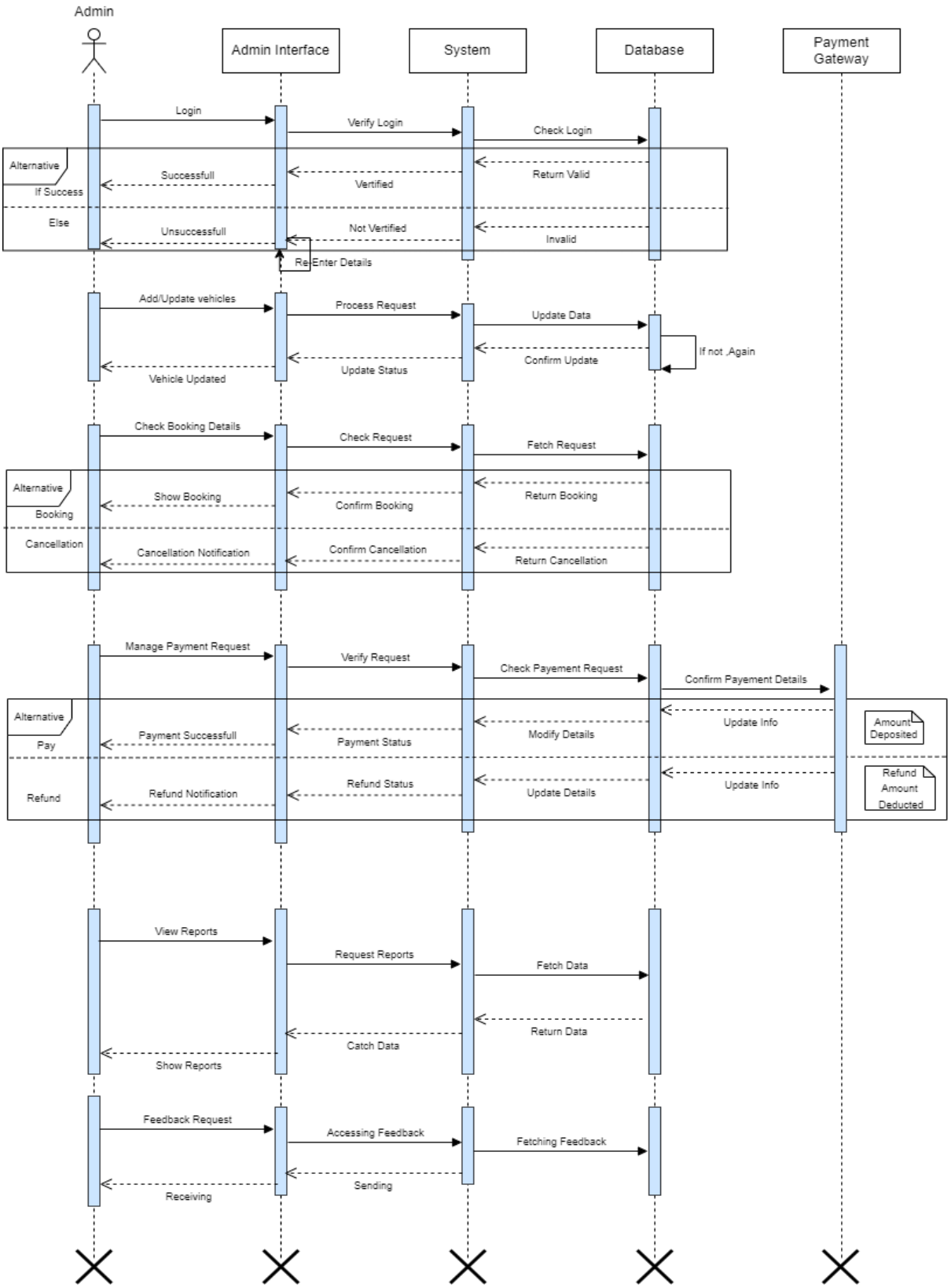

Details
+documentType: String
+document: File
+id :String
User
- userId: int
+ name: String
+ email: String
+ password: String
+ phoneNumber: String
+ address: String
+IDtype:Boolean
+ login(): void
+ logout(): void
Vehicle
- vehicleId: int
+ vehicleType: String
# availability: boolean
+ pricePerDay: double
#gpsenable:boolean
+numberplate:String
+ updateStatus(): void
+ checkAvailability(): Boolean
Admin
+ dashboardAccess: boolean
+ generateReports(): void
+manageVehicles():void
+manageBookings():void
-verifyDocument(): boolean
Customer
- bookingHistory: List<Booking>
+ makeBooking(): void
+ leaveFeedback(): void
+searchVehicles(): void
-uploadDocument():void
Bike
+ bikeType: String
Scooter
+ scooterType: String
Booking
- bookingId: int
+ StartDate: DateTime
+EndDate:DateTime
+ vehicleId: Vehicle
+ IDtype: Boolean
+UserId:int
+ totalAmount: double
+BookingStatus:String
+ confirmBooking(): void
+ cancelBooking(): void
Payment
- paymentId: int
+ amount: double
+ paymentMethod: String
+ bookingid: Booking
#status: String
+ processPayment(): void
+ issueRefund(): void
Report
- reportId: int
- userId: int
-bookingid:int
- feedbackid:int
+ generateReport(): void
+ viewReportDetails(): String
Feedback
- feedbackId: int
+ comments: String
+userid: int
+ submitFeedback(): void
1
0..*

4.1.4 Sequence Diagram

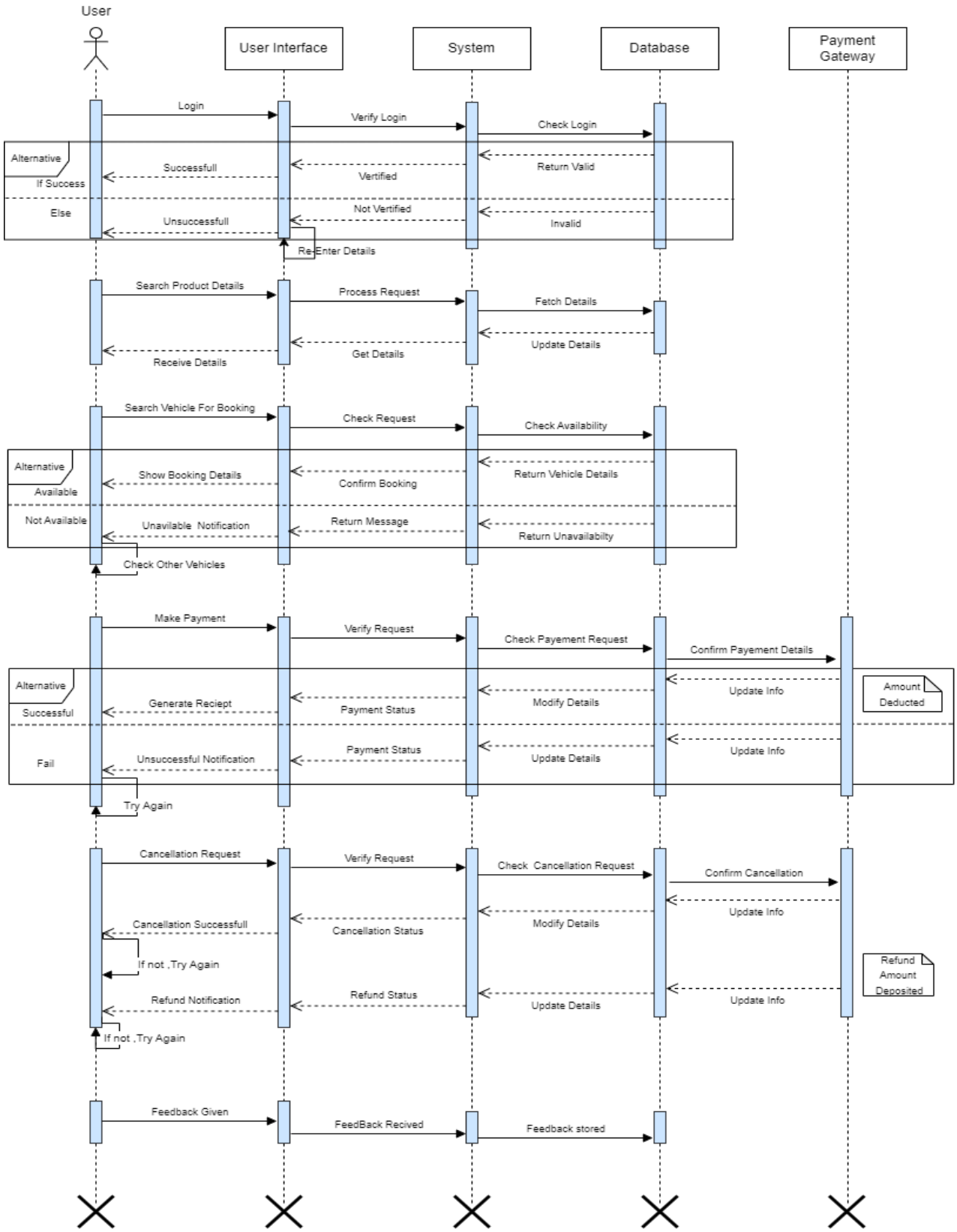
A **sequence diagram** is a type of UML diagram that illustrates how objects interact in a specific scenario of a system. It shows the sequence of messages exchanged between objects over time, helping to visualize the dynamic behavior of a system and the order of operations involved in a particular use case.

Symbol	Description
	Actor
	Object
	Lifeline
	Activation Box
	Message Call
	Return Message
	Note
	Alternatives
	Self Loop
	Death Marker

Admin :

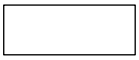
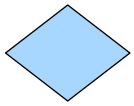

Admin
Admin Interface
System
Database
Payment Gateway
Login
Verify Login
Check Login
Alternative
If Success
Successfull
Vertified
Return Valid
Else
Unsuccessfull
Not Vertified
Invalid
Re-Enter Details
Add/Update vehicles
Process Request
Update Data
If not ,Again
Confirm Update
Update Status
Vehicle Updated
Check Booking Details
Check Request
Fetch Request
Alternative
Booking
Show Booking
Confirm Booking
Return Booking
Cancellation
Cancellation Notification
Confirm Cancellation
Return Cancellation
Manage Payment Request
Verify Request
Check Payement Request
Confirm Payement Details
Alternative
Pay
Payment Successfull
Payment Status
Modify Details
Update Info
Amount Deposited
Refund
Refund Notification
Refund Status
Update Details
Update Info
Refund Amount Deducted
View Reports
Request Reports
Fetch Data
Return Data
Catch Data
Show Reports
Feedback Request
Accessing Feedback
Fetching Feedback
Sending
Receiving

User :

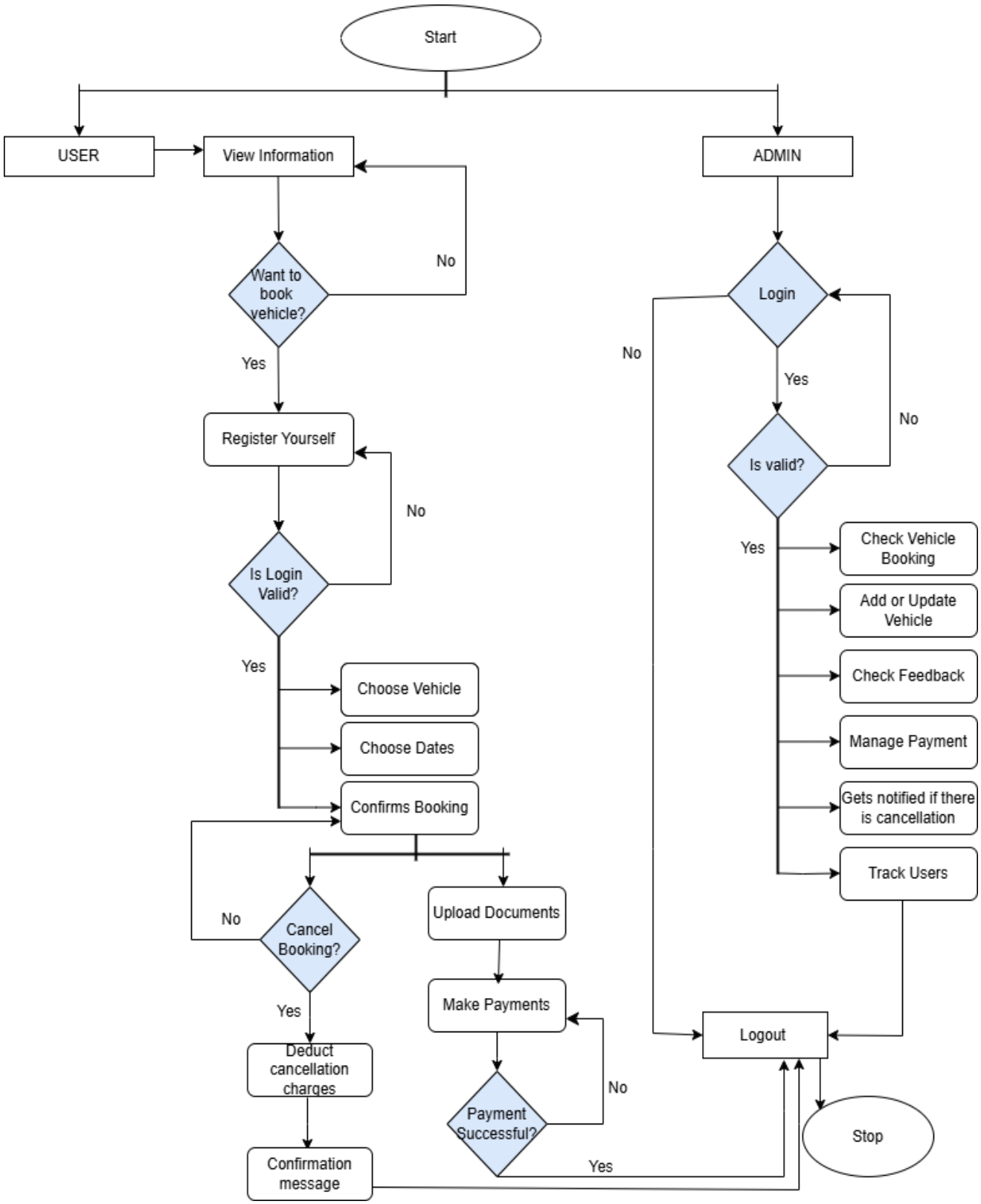

User
User Interface
System
Database
Payment Gateway
Login
Verify Login
Check Login
Alternative
If Success
Successfull
Vertified
Return Valid
Else
Not Vertified
Unsuccessfull
Invalid
Re-Enter Details
Search Product Details
Process Request
Fetch Details
Update Details
Get Details
Receive Details
Search Vehicle For Booking
Check Request
Check Availability
Alternative
Available
Show Booking Details
Confirm Booking
Return Vehicle Details
Not Available
Unavilable Notification
Return Message
Return Unavailabilty
Check Other Vehicles
Make Payment
Verify Request
Check Payement Request
Confirm Payement Details
Alternative
Successful
Generate Reciept
Payment Status
Modify Details
Update Info
Amount Deducted
Fail
Unsuccessful Notification
Payment Status
Update Details
Update Info
Try Again
Cancellation Request
Verify Request
Check Cancellation Request
Confirm Cancellation
Update Info
Cancellation Successfull
Cancellation Status
Modify Details
If not ,Try Again
Refund Amount Deposited
Refund Notification
Refund Status
Update Details
Update Info
If not ,Try Again
Feedback Given
FeedBack Recived
Feedback stored

4.2 System Flow Diagram

A **system flow diagram** visually represents the flow of data and processes within a system. It shows how inputs are transformed into outputs through various processes, helping to clarify the interactions and overall functionality of the system.

Symbol	Description
	Start Node
	Activity Node
	Decision Node
	Control Flow
	Stop Node

System :


Start
USER
View Information
ADMIN
Want to book vehicle?
No
Yes
Register Yourself
No
Is Login Valid?
Yes
Choose Vehicle
Choose Dates
Confirms Booking
No
Cancel Booking?
Yes
Deduct cancellation charges
Confirmation message
Upload Documents
Make Payments
Payment Successful?
No
Yes
Login
No
Yes
No
Is valid?
Yes
Check Vehicle Booking
Add or Update Vehicle
Check Feedback
Manage Payment
Gets notified if there is cancellation
Track Users
Logout
Stop

o

4.3 Data Dictionary

A **data dictionary** is a centralized repository that defines and describes the data elements in a system or database. It includes details such as data types, formats, relationships, and constraints, serving as a reference for developers and users to ensure consistent data usage and understanding.

User Table

Attribute Name	Data Type	Description	Constraint
user_id	int	Unique identifier for each user	Primary Key, Auto-generated
Id_type	Boolean	Specifies whether he is user or admin	(True:Admin False:User) Default:False
name	String	Full name of the user	Not null
email	String	Email of the user	Not null, unique
password	String	Hashed password for secure user login	Not null
Phone	String	User's contact number	Not null
Address_id	Int	Reference to the address table	Foreign key(address table)
Details_id	Int	Reference to the detail table	Foreign key(detail table)

Address Table

Attribute	Data Type	Description	Constraint
Address_id	Int	Unique identifier for the address	Primary key,Auto-generated
Street	String	Street name and number	Not null
City	String	City of the user	Not null
State	String	State of the user	Not null
PinCode	String	Pincode of the user	Not null

Vehicles Table

Attribute Name	Data Type	Description	Constraint
vehicle_id	int	Unique identifier for each vehicle	Primary Key, Auto-generated
Vehicle_type	String	Type of vehicle (e.g., bike, scooter)	Not null
Available status	Boolean	Check Availability	Default: True
price_per_day	Decimal	Rental rate per day	Not null
Gps_enabled	Boolean	Indicates if GPS is installed on the vehicle	Default: False
Number_plate	String	Number plate of the vehicle	Not null, unique

Booking_Table

Attribute Name	Data Type	Description	Constraint
Booking_id	Int	Unique identifier for each booking	Primary-key, Auto-generated
vehicle_id	Int	Reference to the booked vehicle	Foreign Key (Vehicles table), Not Null
Id_type	Boolean	Customer making the booking	Foreign Key (users table), Not Null
User_id	Int	Customer making the booking	Foreign Key (users table), Not Null
Start_date	DateTime	Rental Start date	Not null
End_date	DateTime	Rental end Date	Not null
Booking status	String	Status of the booking (pending, confirmed, cancelled)	Default: "pending"
Total_amount	Decimal	Total rental cost for the booking	Auto-calculated, Not Null

Payment Table

Attribute Name	Data Type	Description	Constraint
payment_id	int	Unique identifier for each payment	Primary Key, Auto-generated
Booking_id	int	Reference to the associated booking	Foreign Key (Bookings table), Not Null
Payment_method	String	Payment method (e.g., credit card, UPI, etc.)	Not null
amount	Double	Payment amount	Not null
Payment_status	String	Payment status (paid, pending, failed)	Default: "pending"

Report Table

Attribute	Data Type	Description	Constraints
reportID	int	Unique identifier for each report	Primary Key,Auto-generated
User_id	Int	Admin who is viewing the report	Foreign Key (Users Table)
booking_id	Int	To track information of users	Foreign key (Bookings table)
Feedback_id	Int	To check which users have given feedback	Foreign key (Feedback table)

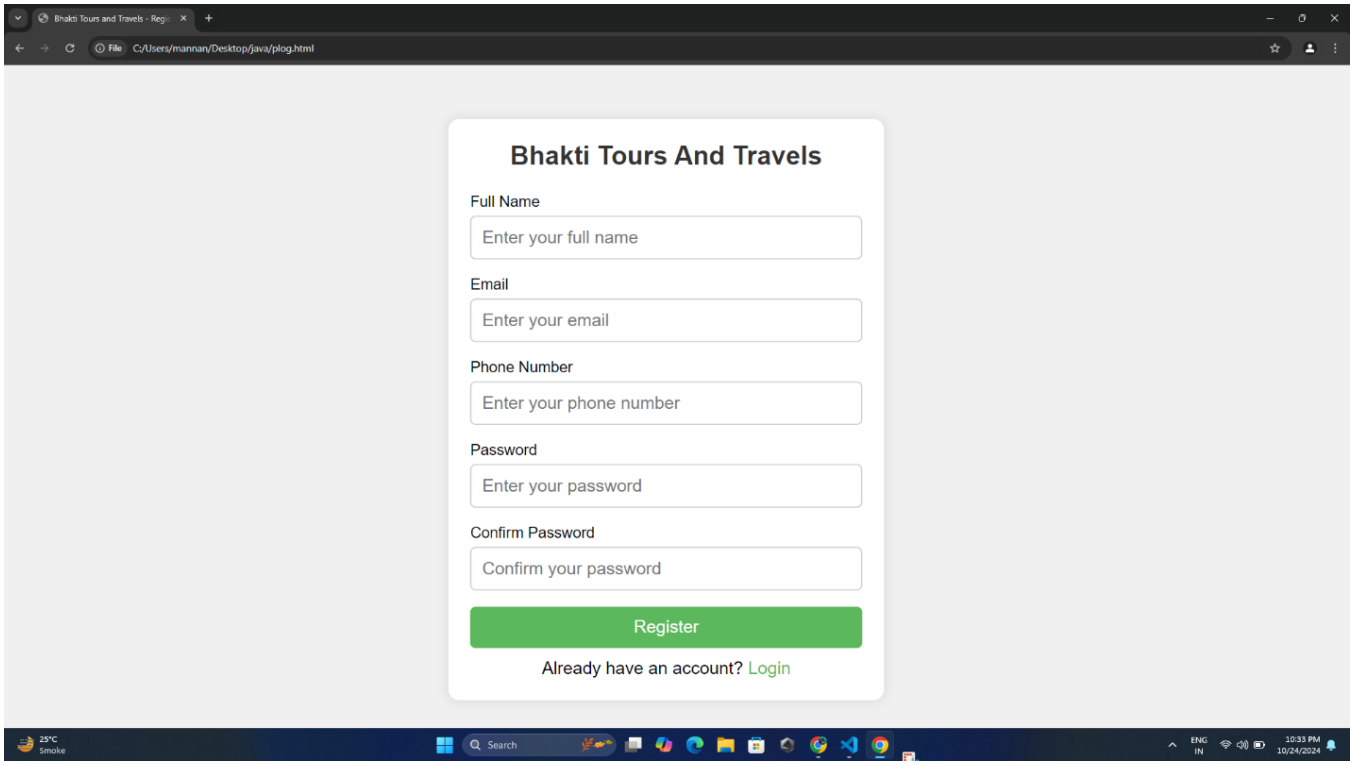
Feedback Table

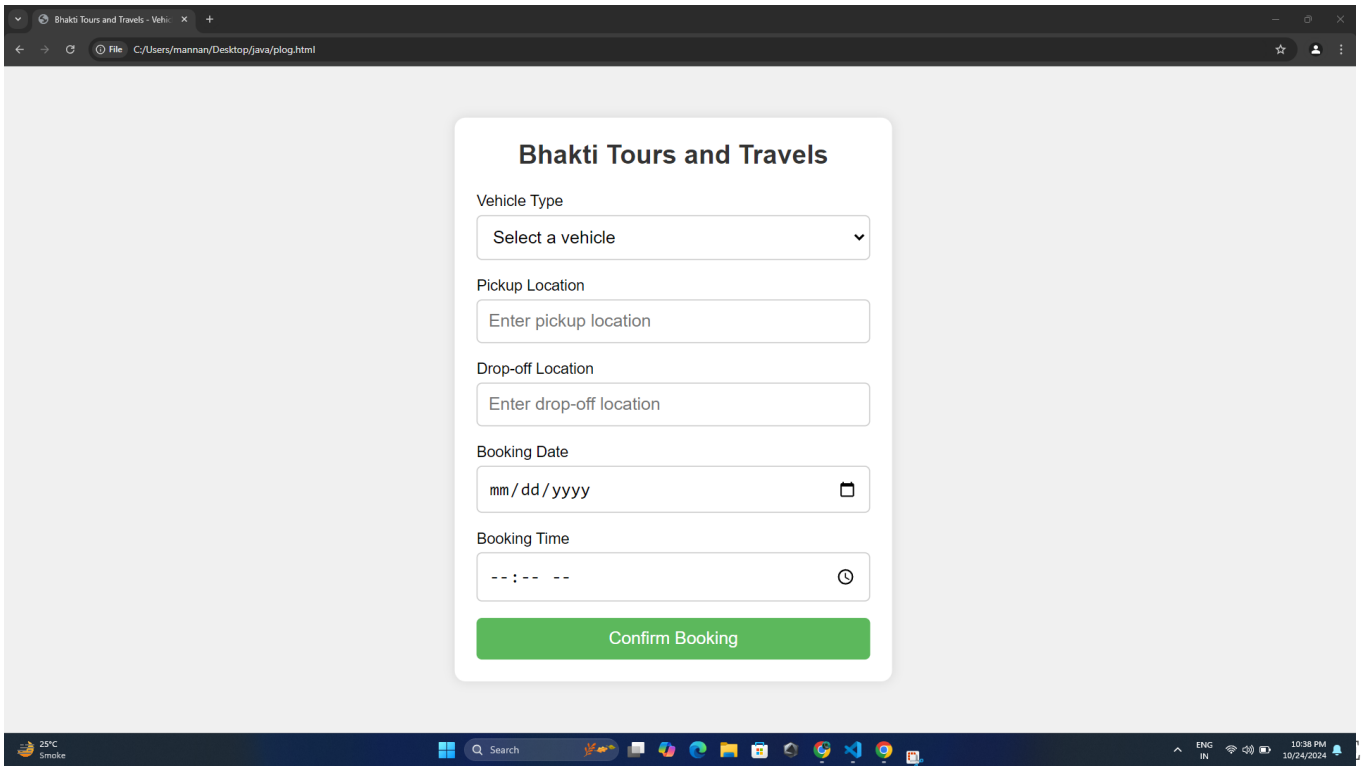
Attribute Name	Data Type	Description	Constraint
feedback_id	Int	Unique identifier for each payment	Primary Key, Auto-generated
user_id	Int	Reference to the user providing feedback	Foreign Key (Users table), Not Null
Comments	String	User's comments or feedback	Optional

Details Table

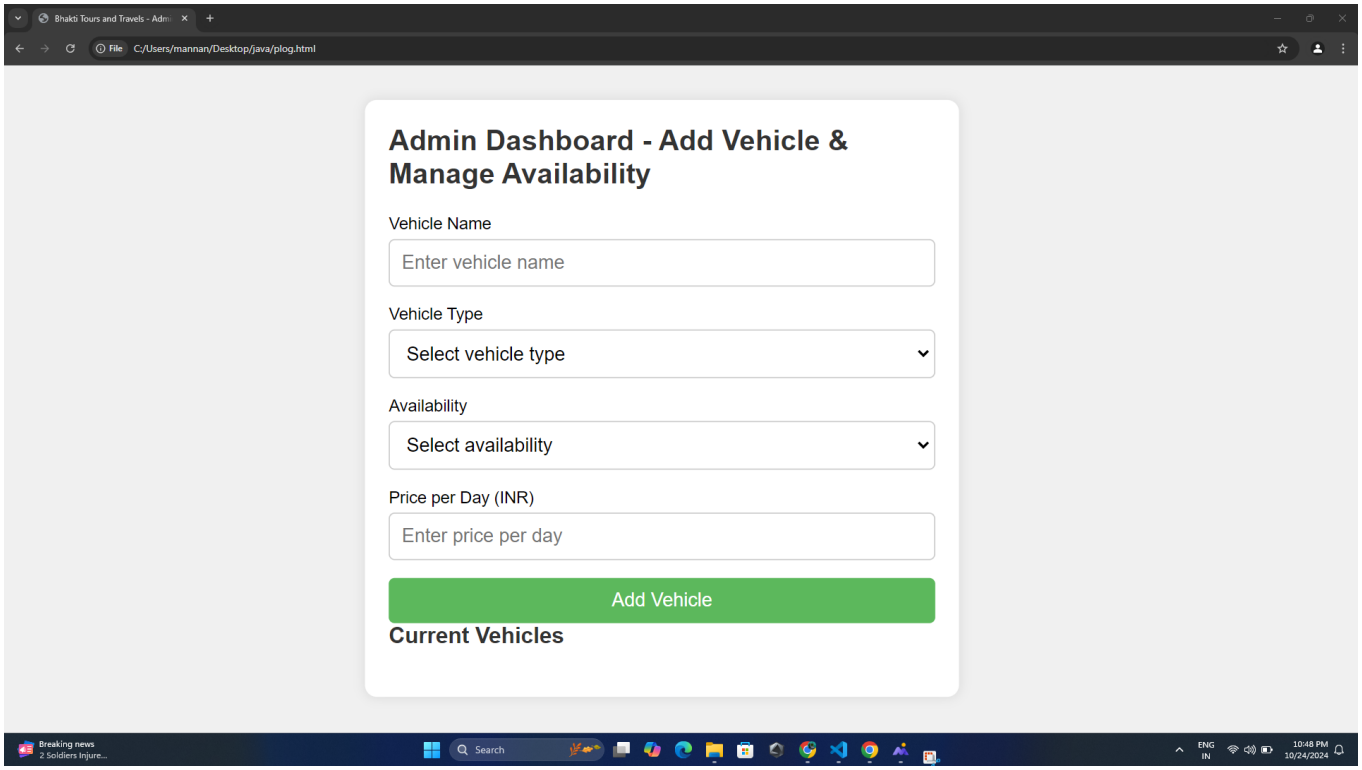
Attribute	Data Type	Description	Constraint
details_id	Int	Unique identifier for each document detail	Primary key,Auto-generated
Document_type	String	Type of the document (e.g., Driving license)	Not null
document	Varchar	The document file (e.g., file path or blob)	Not null
User_id	String	Reference to the user who uploaded the document	Foreign Key (Users table), Not null

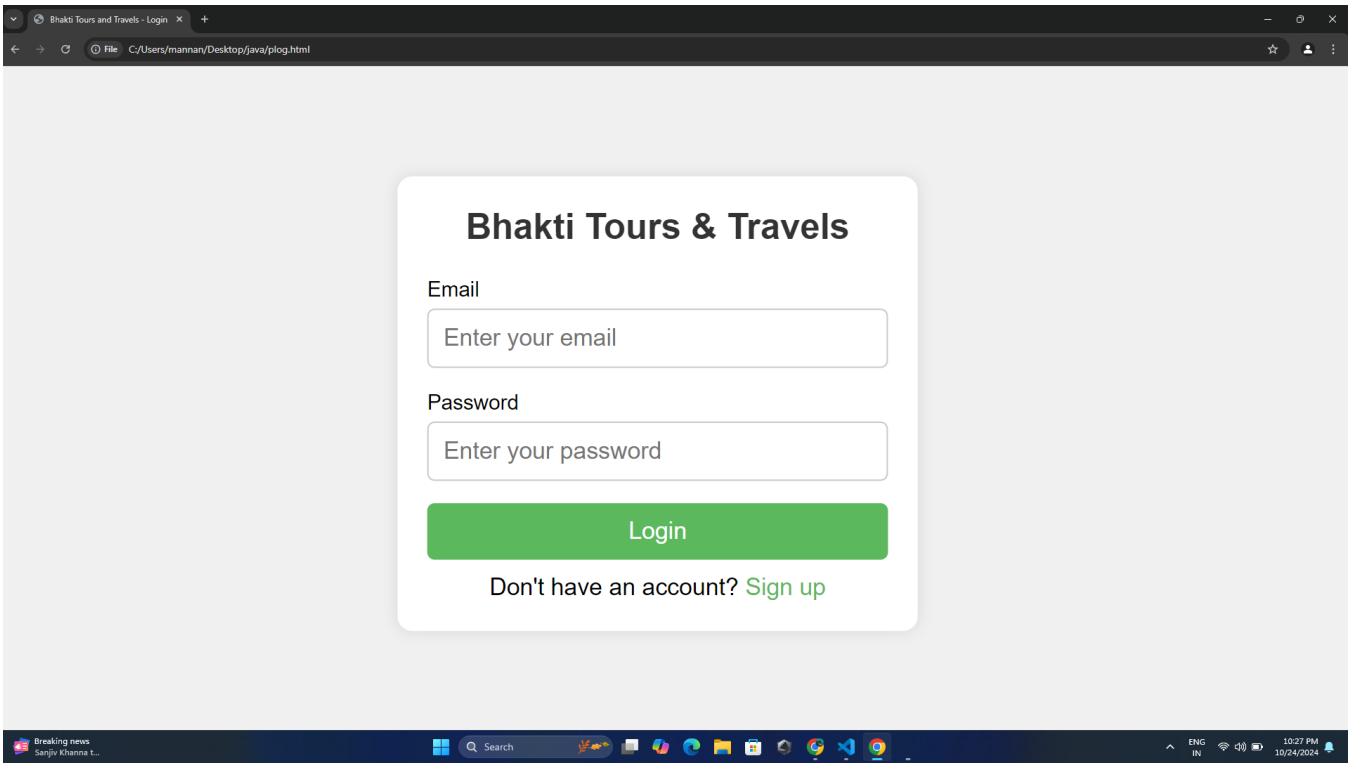
4.4 User Interface

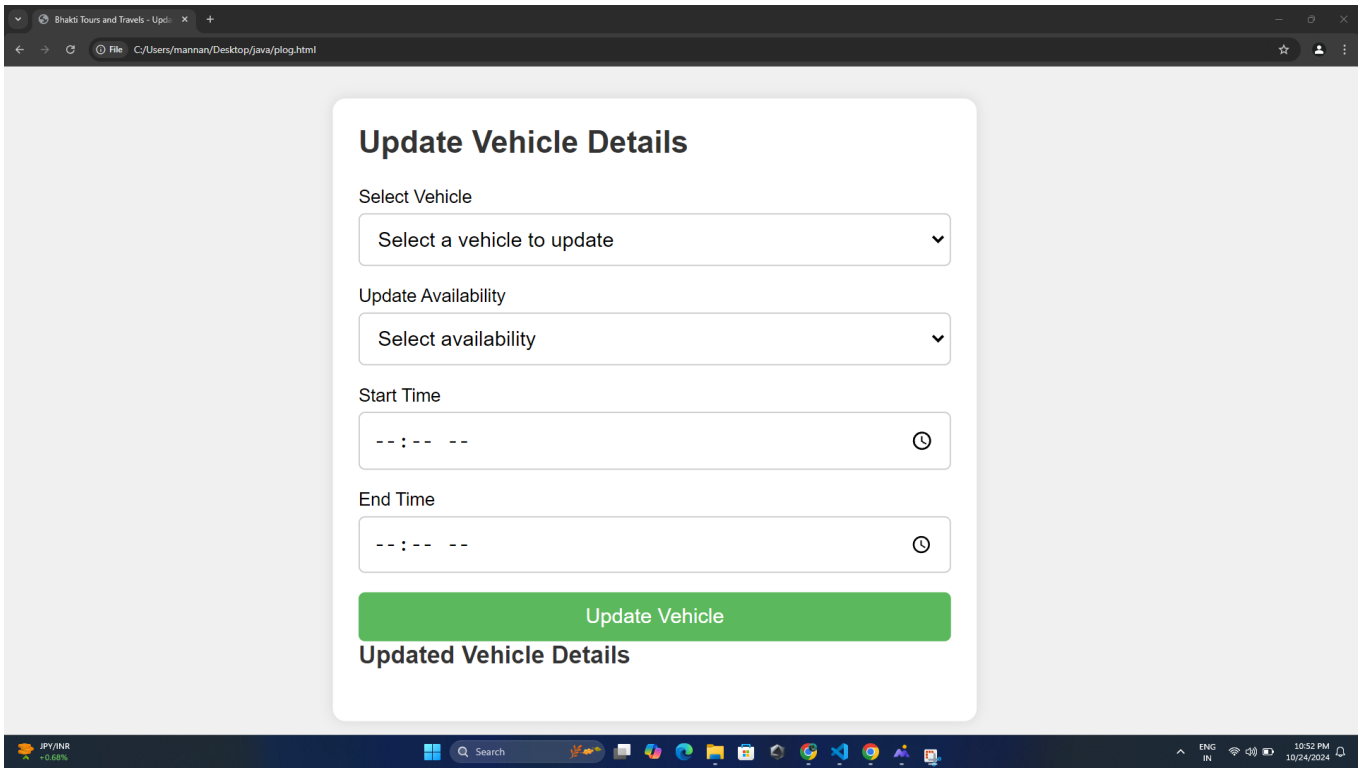


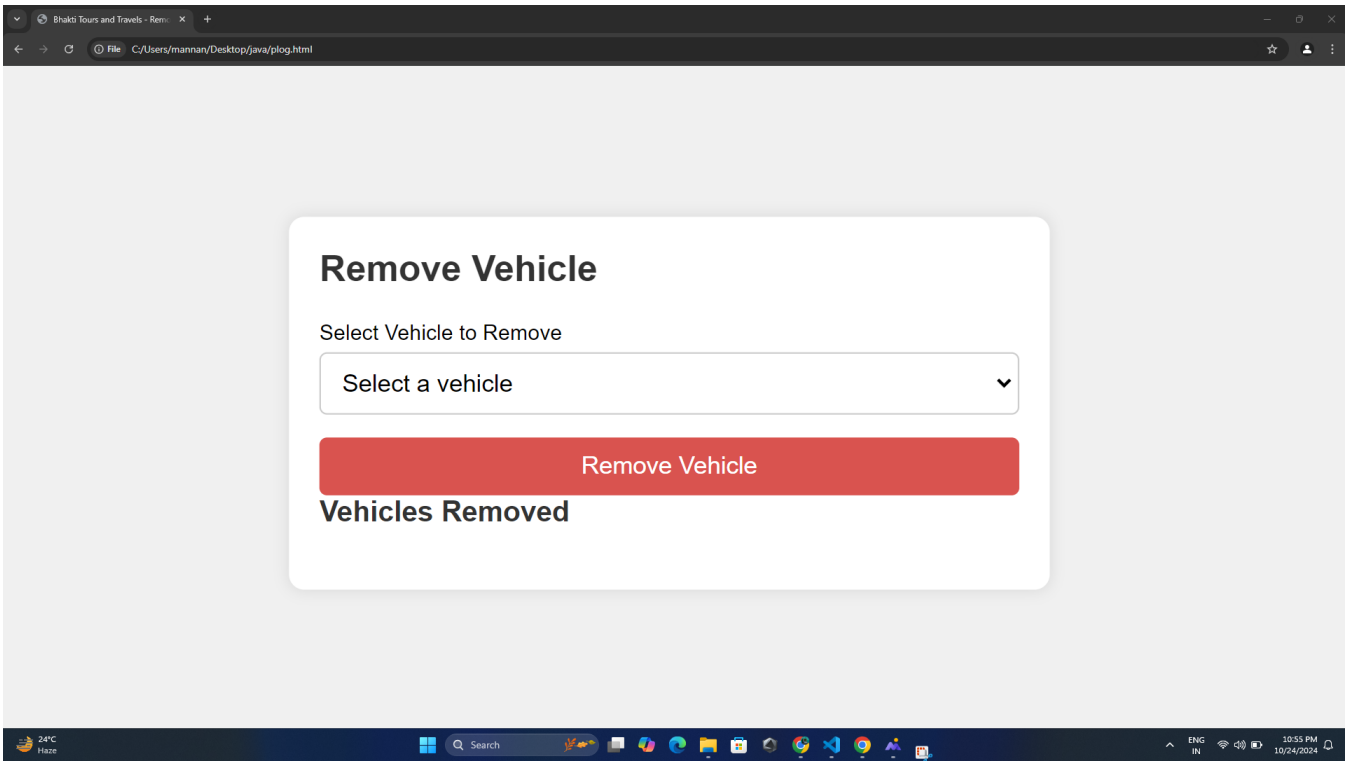


)

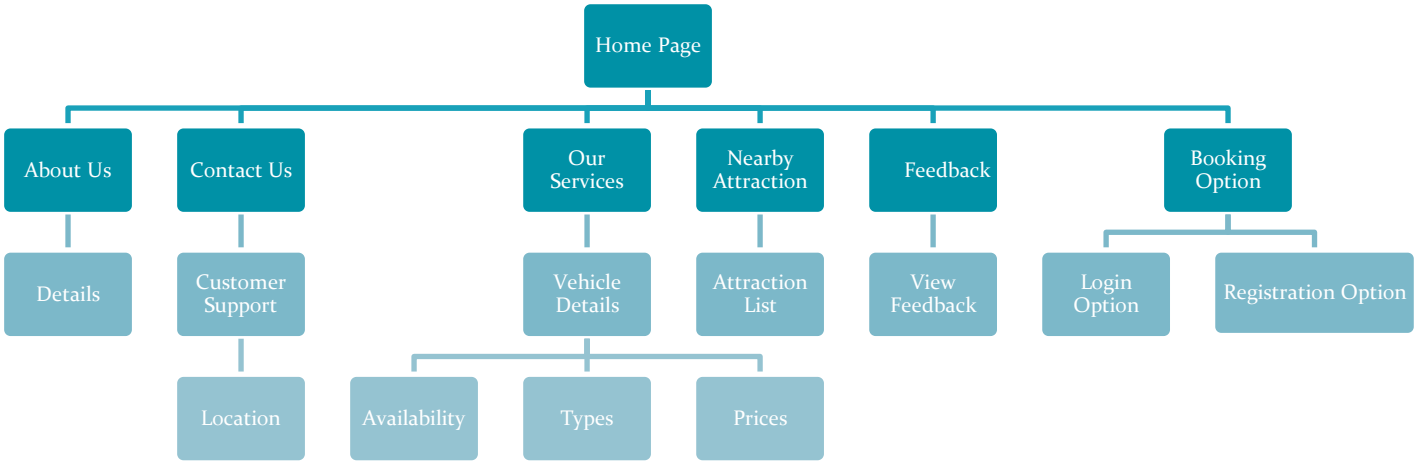

Admin Dashboard - Add Vehicle & Manage Availability
Vehicle Name
Enter vehicle name
Vehicle Type
Select vehicle type
Availability
Select availability
Price per Day (INR)
Enter price per day
Add Vehicle
Current Vehicles


Bhakti Tours & Travels
Email
Enter your email
Password
Enter your password
Login
Don't have an account? Sign up


Bhakti Tours and Travels - Upda
C:/Users/mannan/Desktop/java/plog.html
Update Vehicle Details
Select Vehicle
Select a vehicle to update
Update Availability
Select availability
Start Time
--:-- --
End Time
--:-- --
Update Vehicle
Updated Vehicle Details

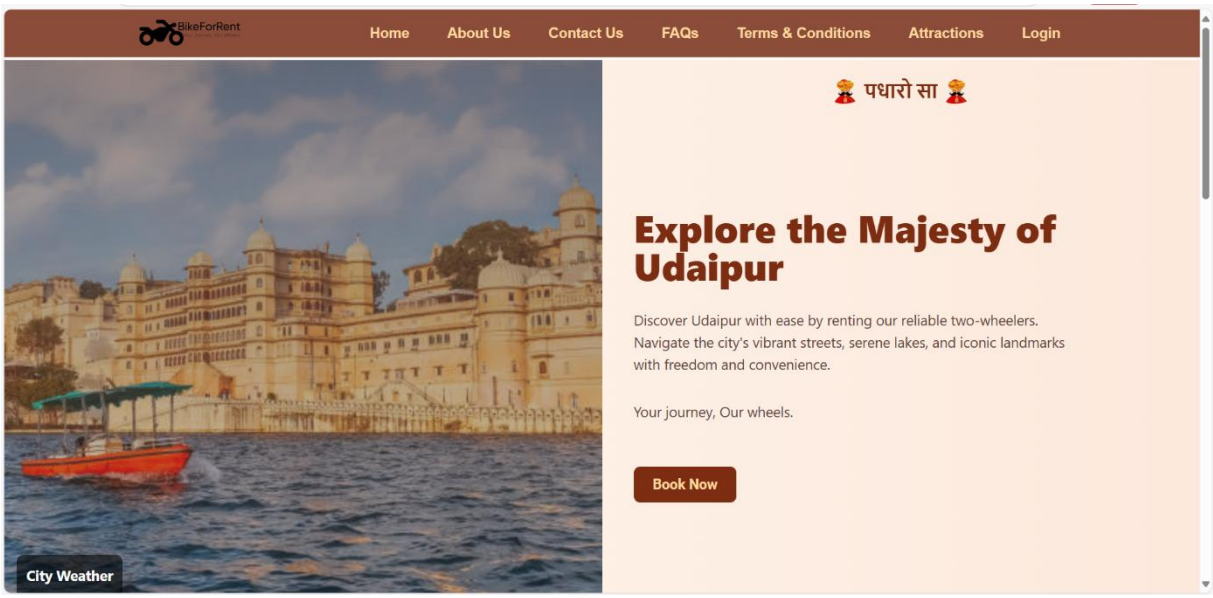

Bhakti Tours and Travels - Remo
C:/Users/mannan/Desktop/java/plog.html
Remove Vehicle
Select Vehicle to Remove
Select a vehicle
Remove Vehicle
Vehicles Removed

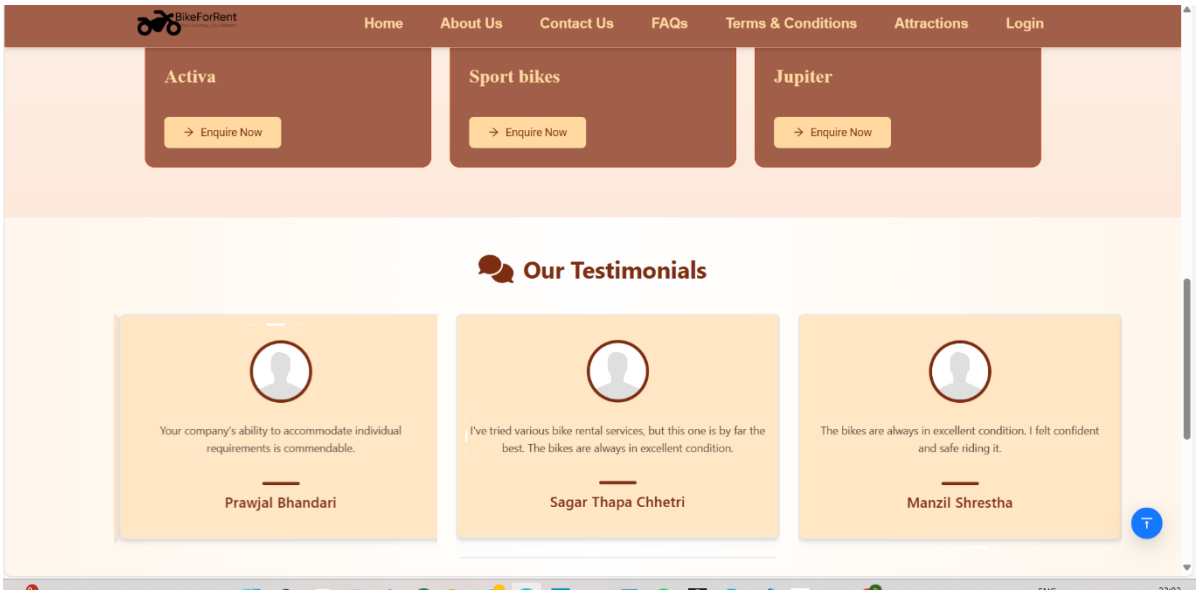
4.5 System Navigation

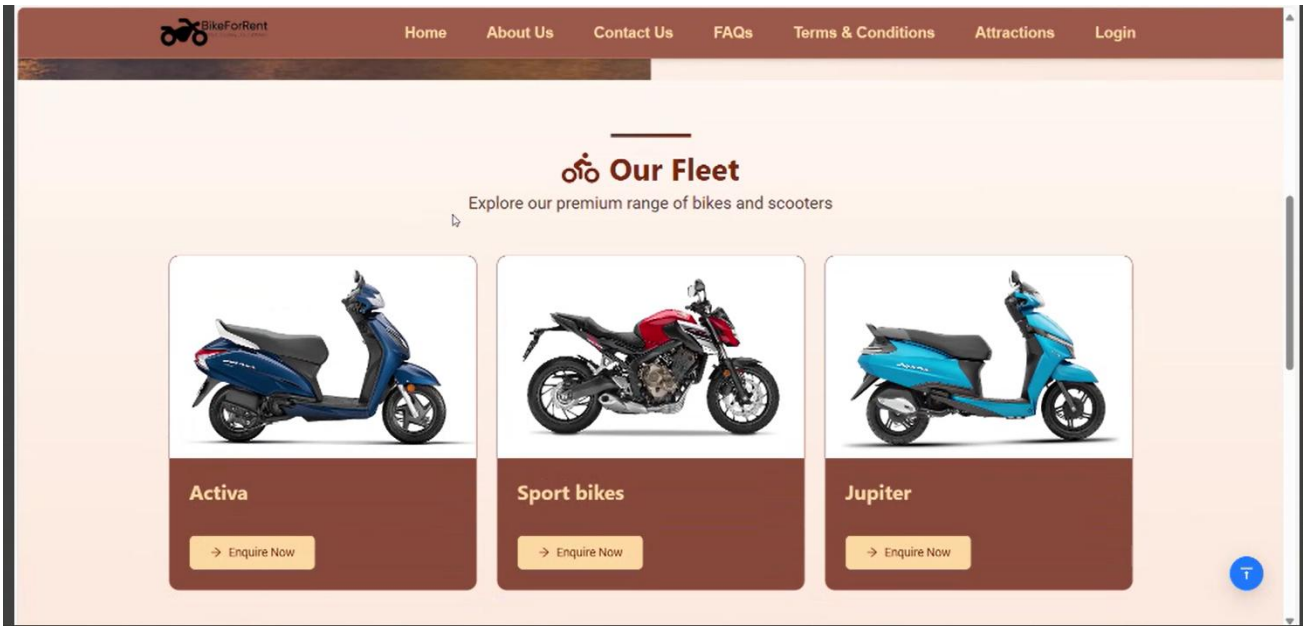


Chapter-5
Input/Output Design

Home Page:
The home page features a navigation bar with options like About Us, Nearby Attractions, and more. It highlights available rental vehicles and provides quick access to booking and other key features,as well as brochure catalog of bikes and testimonials.

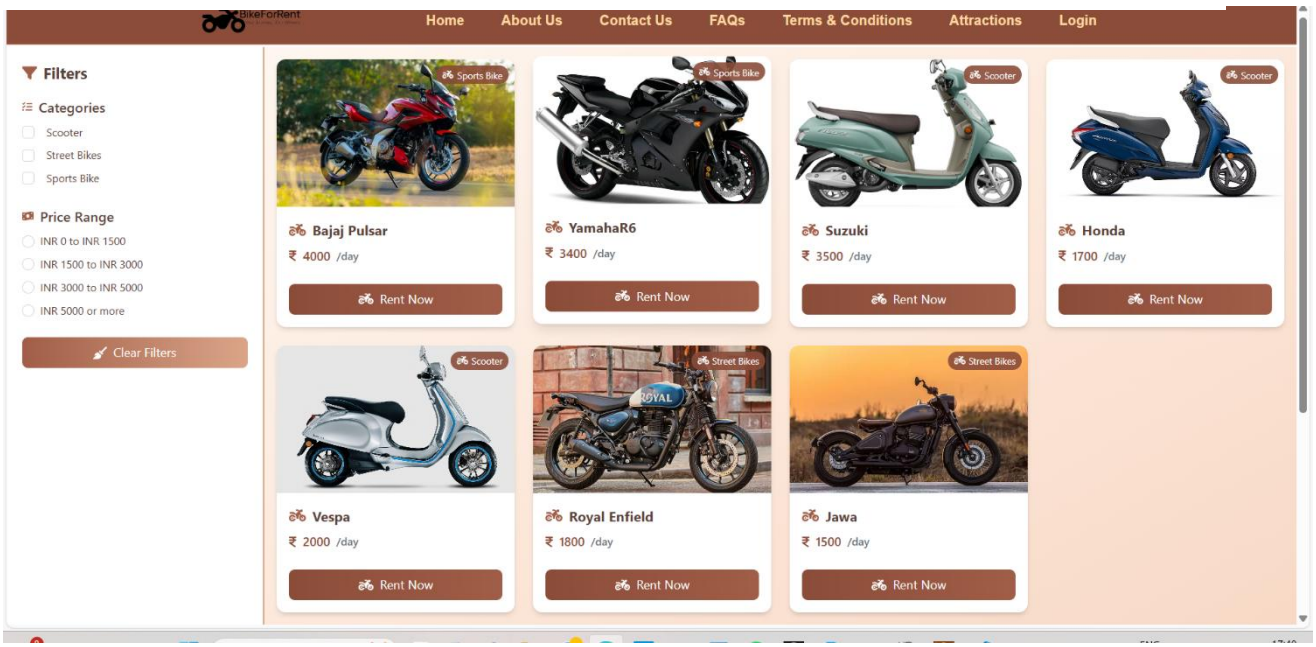


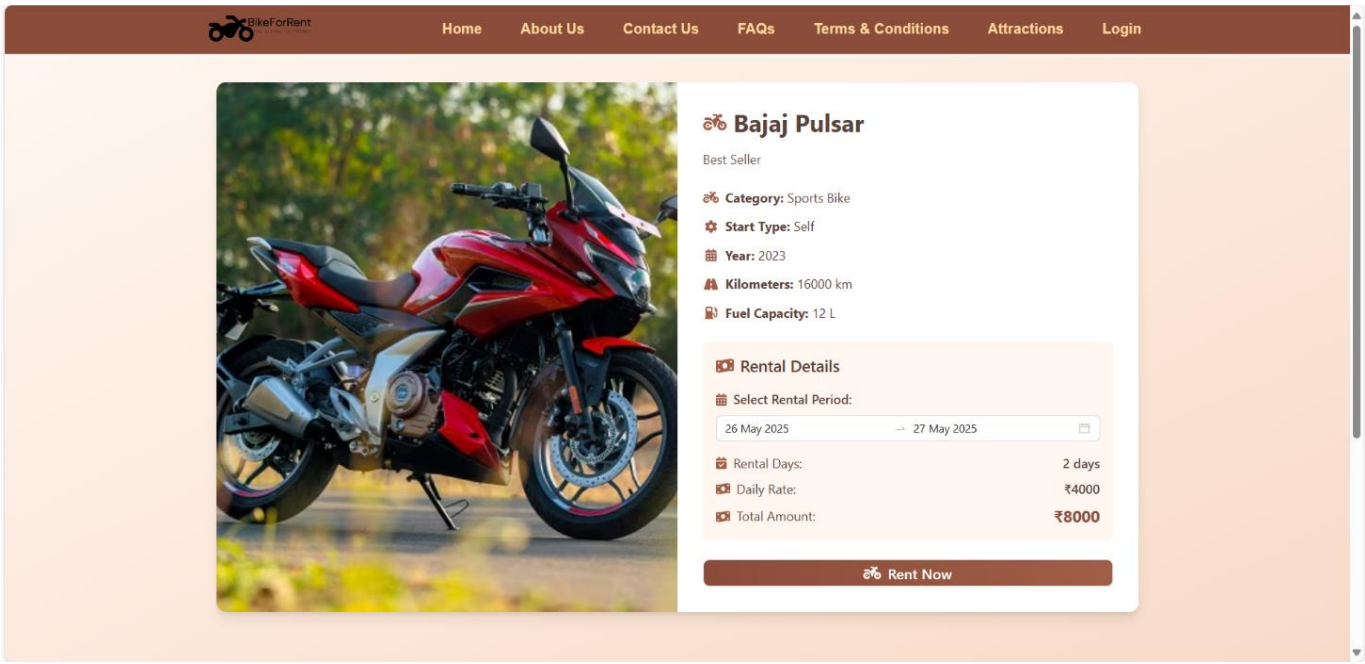


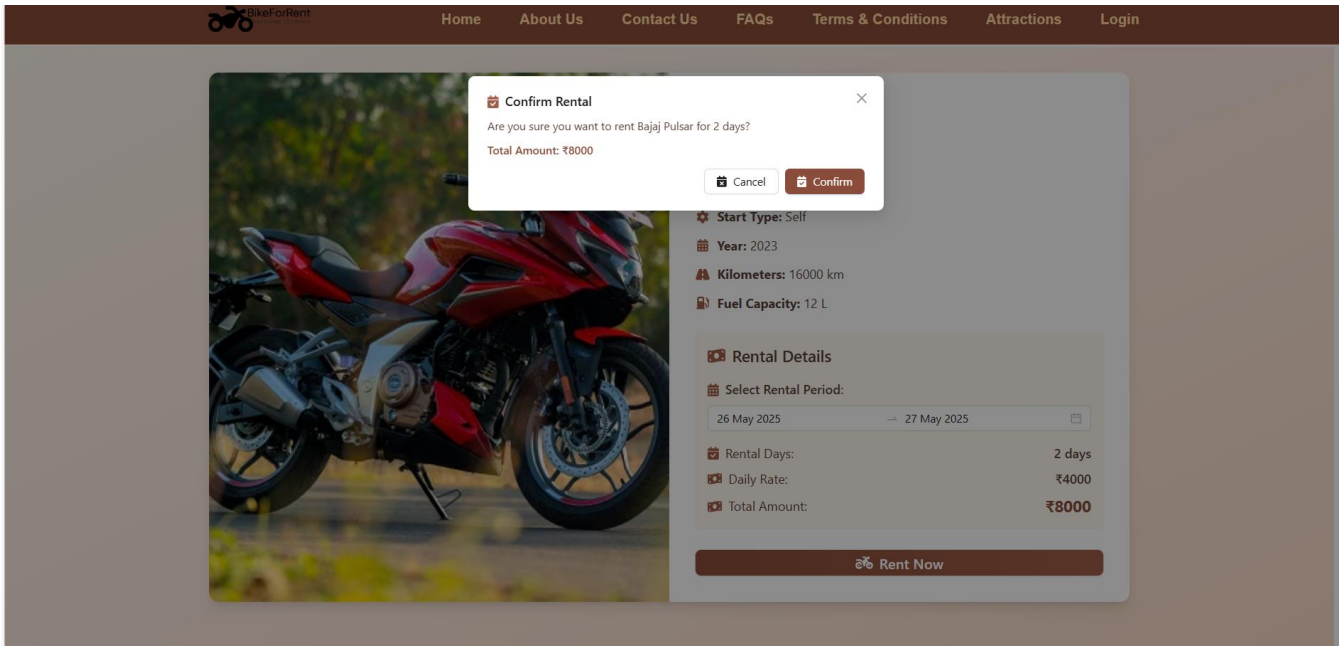


Book Now Page:

The Book Now page displays a variety of rental vehicles, including bikes and scooters, along with their prices and key details. Users can filter options, view availability, and select the vehicle that suits their needs.

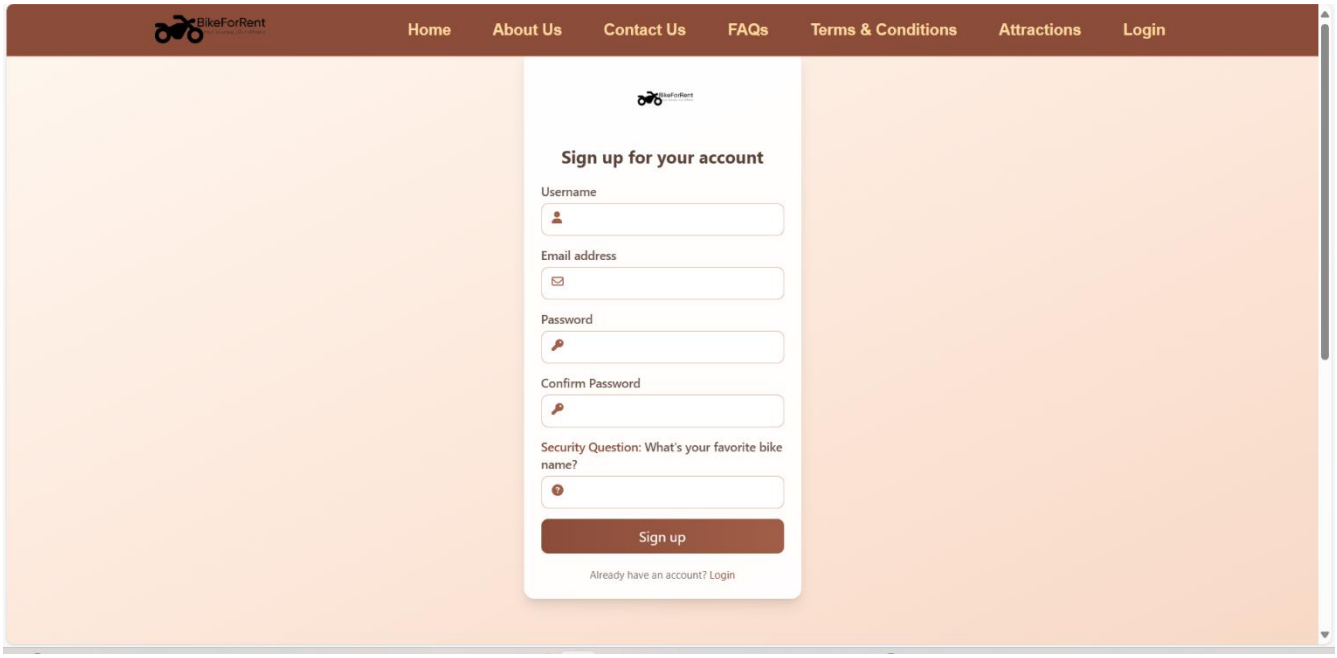


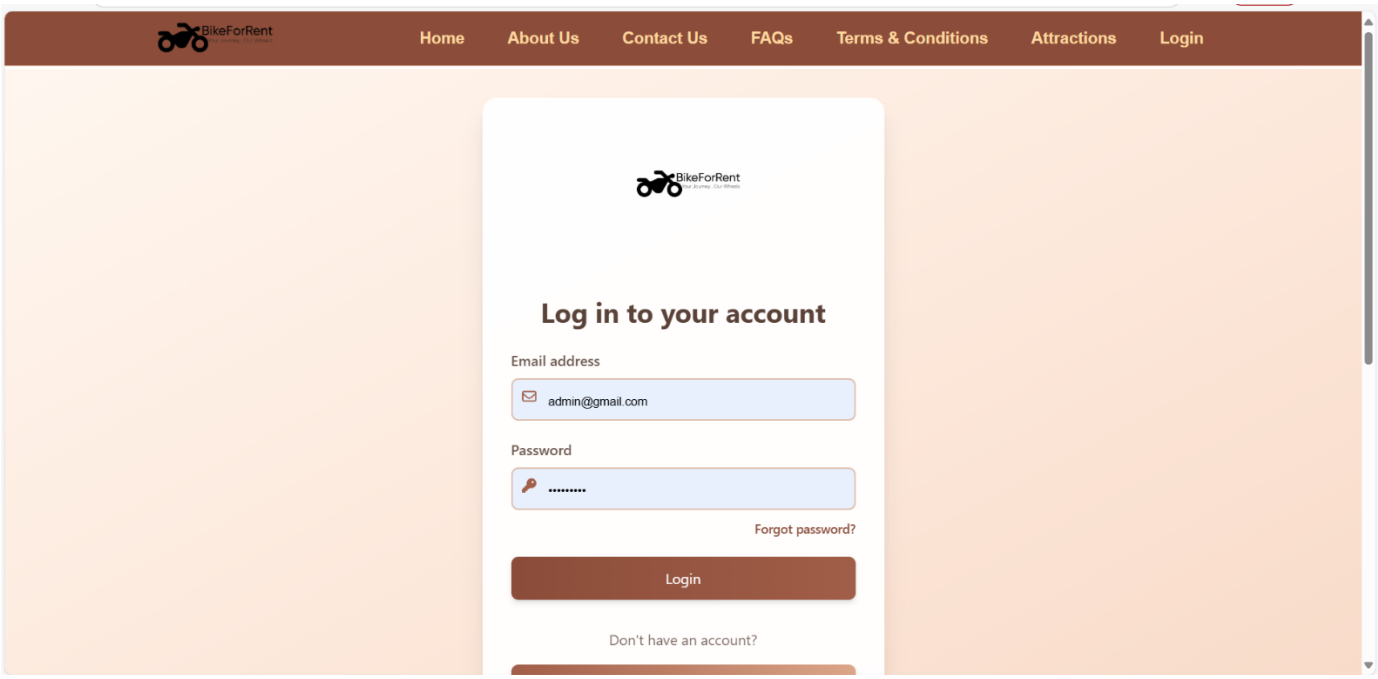

BikeForRent
Home
About Us
Contact Us
FAQs
Terms & Conditions
Attractions
Login
Bajaj Pulsar
Best Seller
Category: Sports Bike
Start Type: Self
Year: 2023
Kilometers: 16000 km
Fuel Capacity: 12 L
Rental Details
Select Rental Period:
26 May 2025
27 May 2025
Rental Days: 2 days
Daily Rate: ₹4000
Total Amount: ₹8000
Rent Now


BikeForRent
Home
About Us
Contact Us
FAQs
Terms & Conditions
Attractions
Login
Confirm Rental
Are you sure you want to rent Bajaj Pulsar for 2 days?
Total Amount: ₹8000
Cancel
Confirm
Start Type: Self
Year: 2023
Kilometers: 16000 km
Fuel Capacity: 12 L
Rental Details
Select Rental Period:
26 May 2025
27 May 2025
Rental Days: 2 days
Daily Rate: ₹4000
Total Amount: ₹8000
Rent Now

Sign Up/Login Page

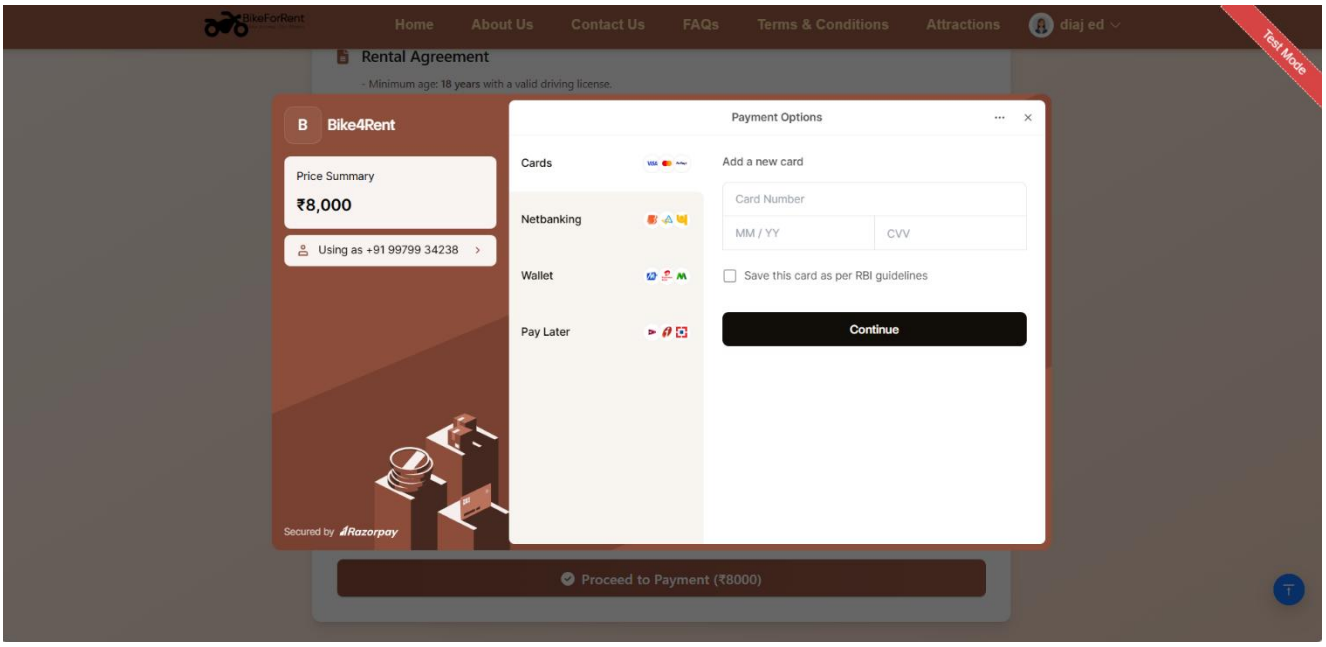
The Sign Up/Login page enables users to create a new account or access their existing one. It provides a simple form for entering credentials, ensuring secure access to booking and payment features.

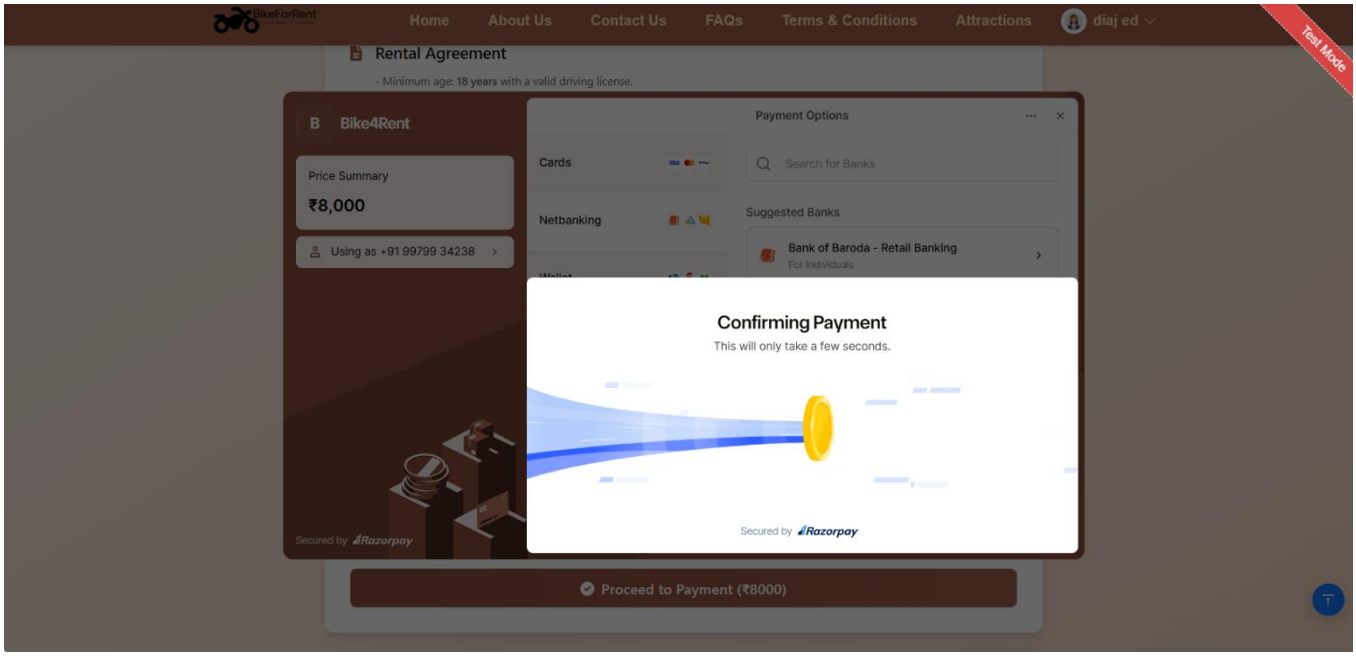




Payment Gateway

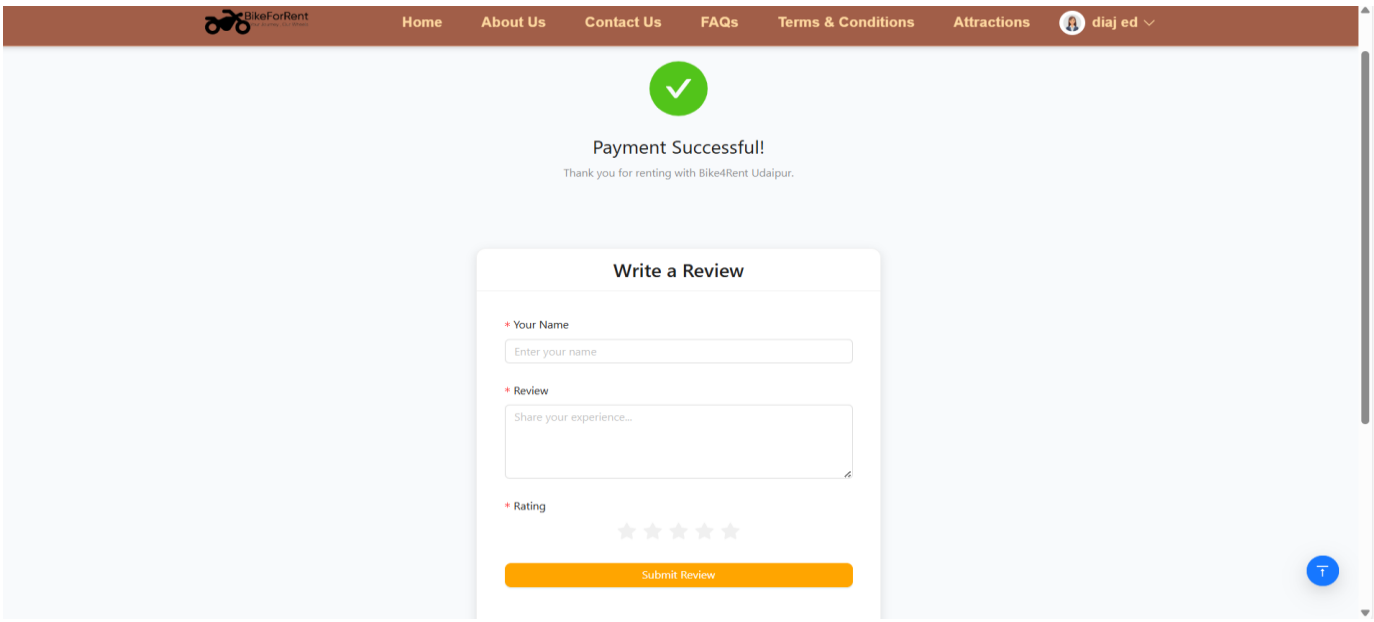
The Payment Gateway page ensures secure transactions, allowing users to complete payments using various methods like credit/debit cards, UPI, or net banking. It confirms successful bookings with a receipt.





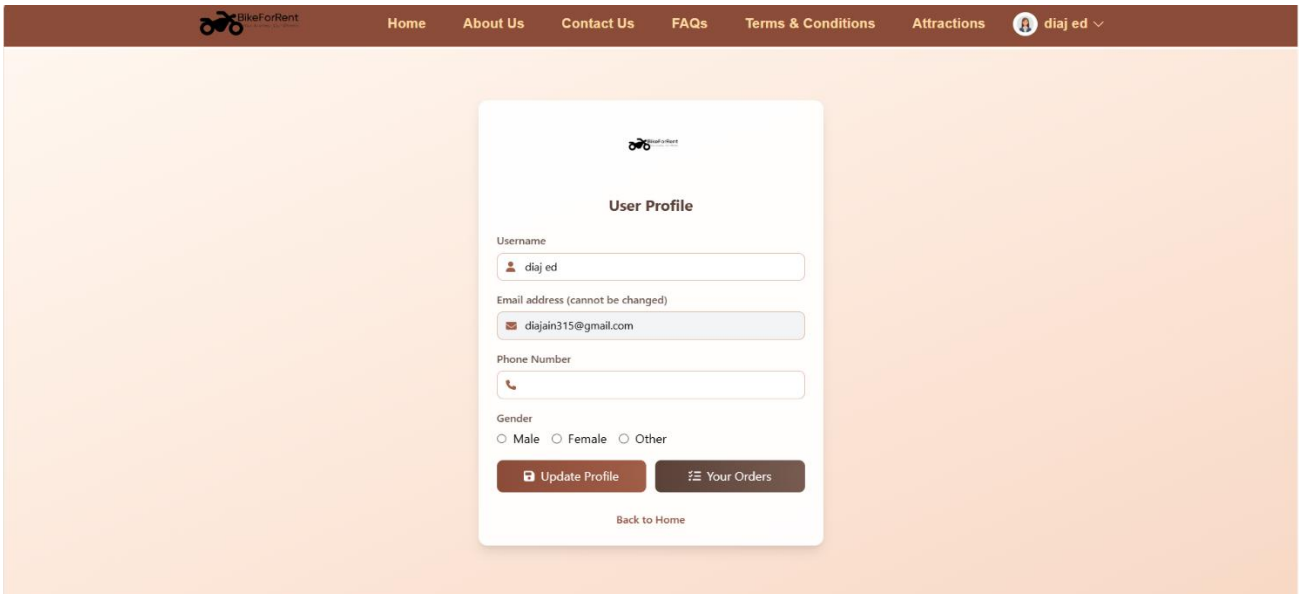
Feedback Page

The Feedback Page allows users to share their experiences and suggestions. It features a simple form to submit ratings and comments, helping improve services.



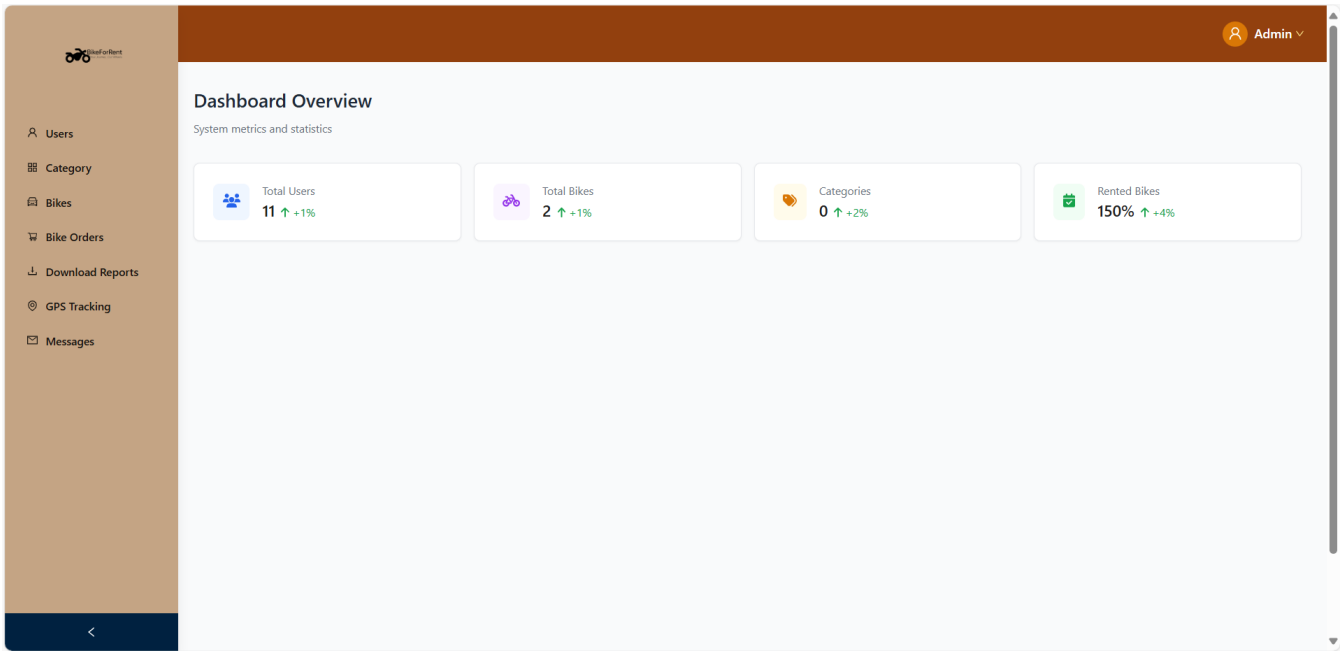
Update Profile Page

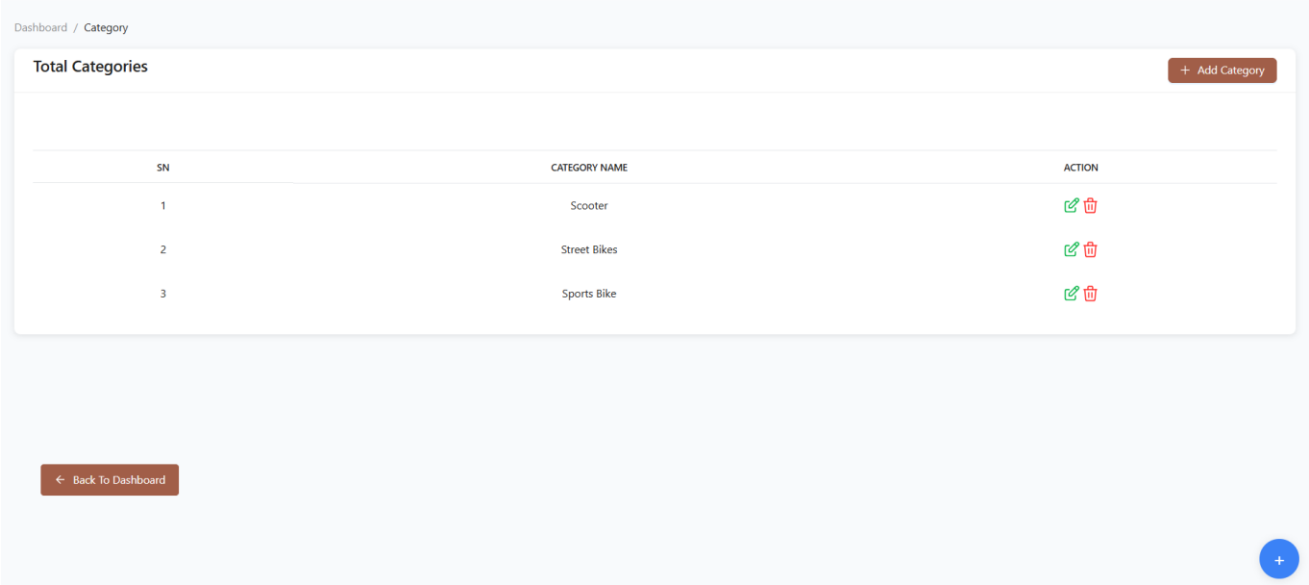
The Update Profile page allows users to edit personal details such as name, contact information, and address. It ensures that the user's profile is always up-to-date for a smooth rental experience.



Admin Interface

The Admin Interface provides tools for managing user accounts, vehicle listings, bookings, and payments. Admins can update vehicle availability, monitor transactions, and handle customer feedback efficiently.





Total Bikes

ADD BIKE

← Back To Dashboard

All Bikes

S.N.	Photo	Name	Number	Price/Day (₹)	Category	Petrol Capacity	Km Used	Start Type	Year	Actions
1		**Bajaj Pulsar**	GJ01WE3456	₹4000	Sports Bike	12	16000	Self	2023	Edit Delete
2		**YamahaR6**	GJ01GH3456	₹3400	Sports Bike	12	15000	Both	2024	Edit Delete
3		**Suzuki**	GJ01ER9012	₹3500	Scooter	12	16900	Self	2023	Edit Delete
4		**Honda**	GJ01RF6785	₹1700	Scooter	14	17000	Self	2012	Edit Delete
5		**Vespa**	GJ01EF1234	₹2000	Scooter	13	19000	Both	2019	Edit Delete
6		**Royal Enfield**	GJ01CD1234	₹1800	Street Bikes	12	16000	Self	2022	Edit Delete

Add New Category

← Back To Categories

* Category Name

e.g., Sccoter

* Upload Category Photo

Click to Upload

Add Category

Order Management

← Back To Dashboard

SN	Bike Name	Bike Number	Rented User	Total Amount	Start Date	End Date	Status
1	Bajaj Pulsar	GJ01WE3456	diaj ed	₹8000	26/05/2025	27/05/2025	Not Processed
2	Bajaj Pulsar	GJ01WE3456	UserA	₹8000	22/04/2025	23/04/2025	Not Processed
3	N/A	N/A	unknown	₹18000	06/06/2025	18/06/2025	Not Processed
4	N/A	N/A	diaj ed	₹1500	15/04/2025	16/04/2025	Not Processed
5	N/A	N/A	diaj ed	₹1200	14/04/2025	15/04/2025	Not Processed
6	N/A	N/A	diaj ed	₹1200	30/04/2025	01/05/2025	Not Processed
7	N/A	N/A	diaj ed	₹1200	28/04/2025	29/04/2025	Not Processed
8	N/A	N/A	diaj ed	₹1500	28/04/2025	29/04/2025	Not Processed
9	N/A	N/A	diaj ed	₹1200	28/04/2025	29/04/2025	Not Processed

Add New Bike

← Back

* Bike Name
e.g., Yamaha FZ

* Bike Number
e.g., GJ01AB1234

* Category
Select Category

* Price Per Day (₹)

* Start Type
Select Start Type

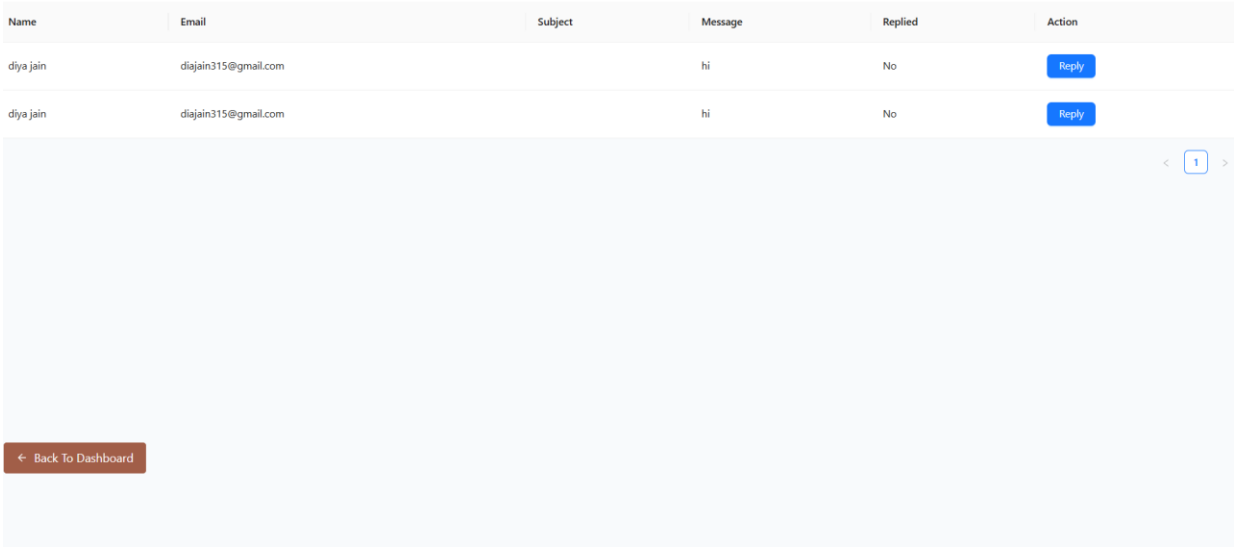
* Year of Manufacture
e.g., 2020

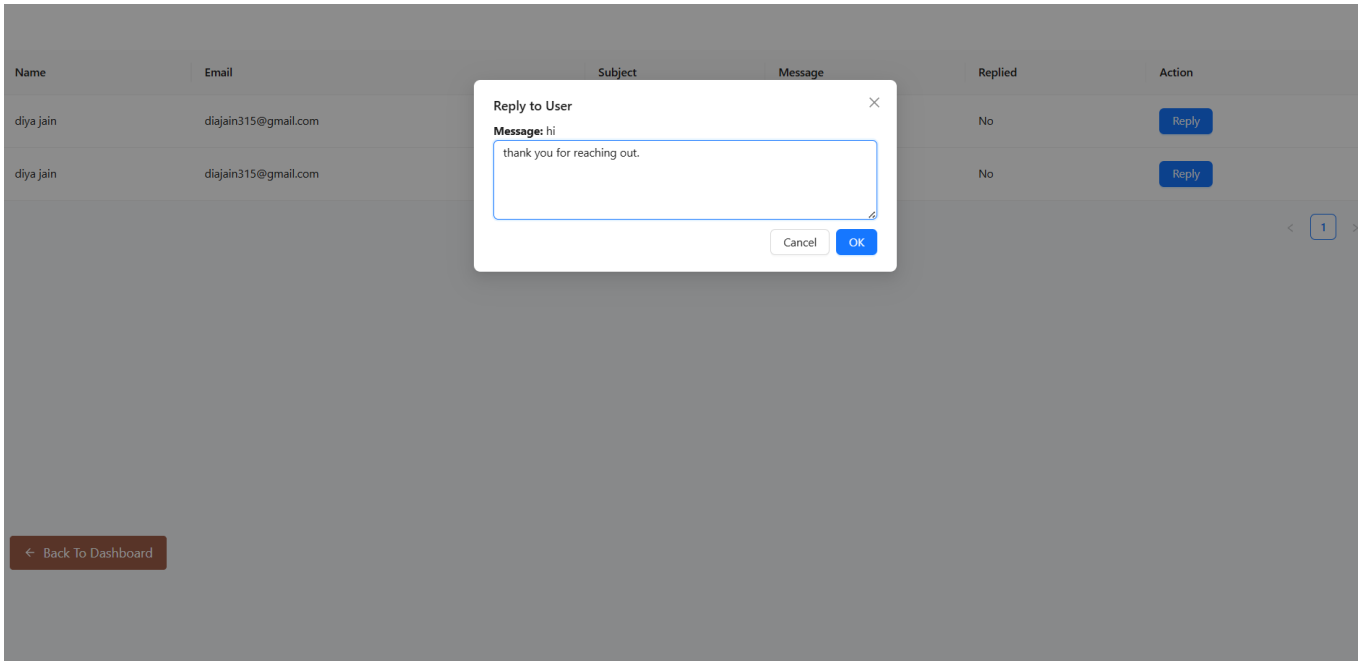
* Kilometers Used
e.g., 15000

* Petrol Capacity (in Litres)
e.g., 12

Messages Page

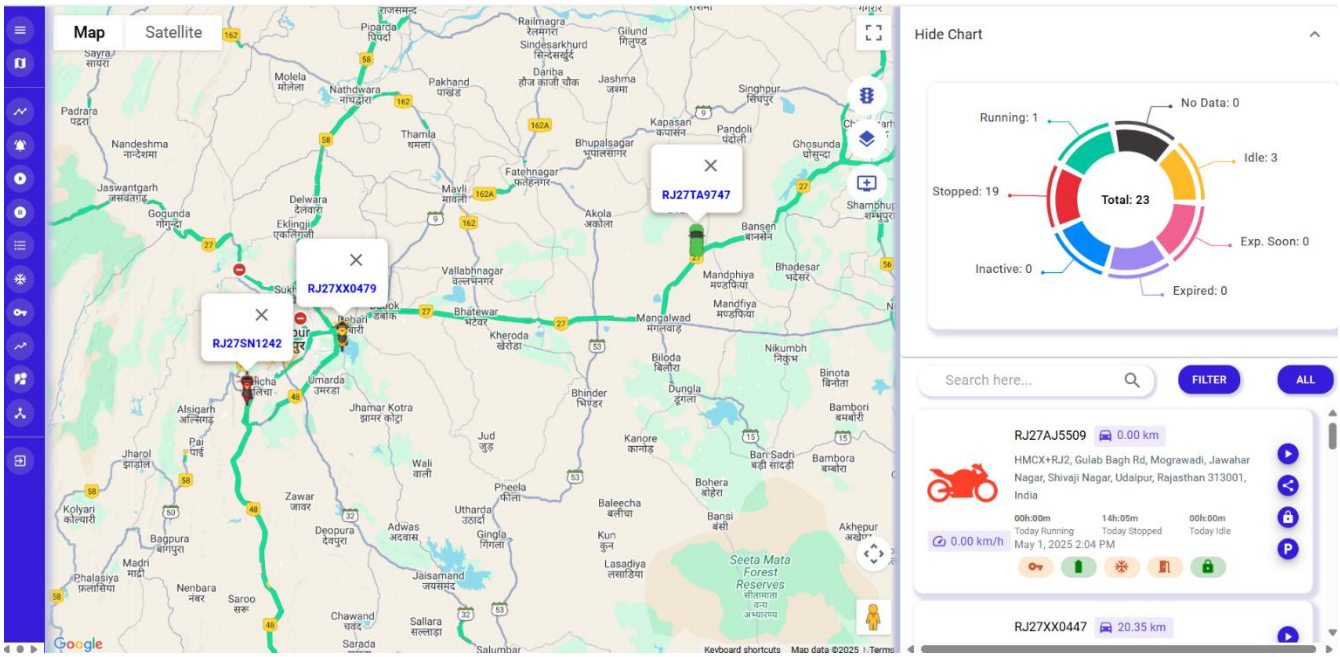
The Messages page allows admins to view and reply to customer inquiries or concerns. It provides a communication channel for resolving issues and offering support to users.





GPS Trekking

The GPS Trekking feature allows users to track their routes and destinations during their treks. It provides real-time navigation and ensures safety by displaying current location and nearby attractions.



Chapter-6
Testing

Test Case ID	Module	Test Scenario	Test Steps	Expected Result	Test Type
TC001	Login/Signup	Verify user can register with valid details	Valid name, email, password	Account created, redirected to homepage	Functional
TC002	Login/Signup	Verify login with valid credentials	Registered email and password	User logged in successfully	Functional
TC003	Login/Signup	Verify login fails with invalid credentials	Wrong email or password	Error message shown	Negative
TC004	Product Listing	Verify all products load correctly on the homepage	Load homepage	List of products displayed	UI/Functional
TC005	Product Details	Verify product details open when clicked	Click on product image/title	Product detail page shown	Functional
TC006	Booking	Verify Booking Vehicle	Click 'Book Now' on a product	Product is booked	Functional
TC007	Checkout	Verify user cannot checkout without login	Add product > Checkout	Prompt to login or register	Security
TC008	Checkout	Verify checkout with valid payment & address	Fill shipping and dummy payment info	Order confirmation shown	Functional
TC009	Order History	Verify order is listed after checkout	Go to 'My Orders'	Recent order listed	Functional
TC010	Search	Verify product search returns correct result	Enter product name in search	Relevant products shown	Functional

TC011	Filters	Verify filter by price/category works	Select price range/category	Filtered results shown	UI/Functional
TC012	Responsive	Verify website on mobile resolution	Open on mobile or simulate	Website fits screen properly	UI/Responsive
TC013	Security	Verify unauthorized access to admin panel is blocked	Visit /admin without login	Redirect to login or error page	Security
TC014	Logout	Verify user is logged out properly	Click on logout	Session ends, redirected to login/homepage	Functional
TC015	UI	Verify all buttons and links work	Click all buttons/links	Correct redirection or action	UI/Functional
TC016	Profile	Update user profile	Navigate to 'My Profile'; edit and save changes	Updated info saved; success message shown	Functional
TC017	Booking	Book a bike with available dates	Select available bike, enter details, confirm booking	Booking confirmed; confirmation message shown	Functional
TC018	Feedback	Submit feedback for a completed ride	Navigate to 'Feedback' post ride; submit rating and comments	Feedback submitted successfully	Functional
TC019	Document Upload	Upload valid document	Go to profile/documents; upload valid ID proof	Document uploaded; status shown	Functional
TC020	Orders	Cancel a pending booking	Go to 'My Bookings'; click 'Cancel' before ride starts	Booking status updated to cancelled	Functional
TC021	Admin Login	Admin login with correct credentials	Enter valid admin email/password	Redirected to admin dashboard	Functional
TC022	Admin Dashboard	View total users, bookings, and revenue	Navigate to admin dashboard	Accurate stats shown	Functional

TC023	Manage Users	Block a user	Select a user from list; click 'Block'	User status changes to blocked; user can't log in	Security
TC024	Manage Vehicles	Add a new vehicle	Fill form with vehicle details, upload image, click 'Add'	Vehicle added; reflected in fleet	Functional
TC025	Manage Vehicles	Edit existing vehicle details	Click 'Edit' on a listed vehicle; update and save changes	Changes reflected in frontend	Functional
TC026	Manage Bookings	View and update booking status	Click on a booking; update status to confirmed or completed	Status updated; user notified	Functional
TC027	Manage Feedback	View user feedback	Navigate to Feedback section	Feedback displayed correctly	Functional
TC028	Admin Logout	Log out from admin panel	Click on 'Logout' in admin panel	Session ends; redirected to login	Functional
TC029	Admin Validation	Ensure only admins can access admin pages	Try to access /admin as user	Access denied or redirect	Security

Chapter-7
Summary

Bhakti Tours and Travels, based in Udaipur, aims to modernize its vehicle rental services by transitioning from a manual, offline system to a comprehensive online platform. The proposed system will streamline operations by introducing features like real-time vehicle availability, secure online bookings, and automated resource management. This user-friendly platform will allow customers to access rental services 24/7, view detailed vehicle information, and make instant bookings. Admins will benefit from a powerful dashboard for managing vehicles, bookings, and customer interactions efficiently.

By leveraging the MERN stack (MongoDB, Express, React, Node.js), the platform will ensure scalability, security, and seamless integration with future enhancements. Additional features, such as GPS tracking and feedback mechanisms, will enhance both user convenience and service quality. This project aligns with the company's vision of providing a smooth, accessible, and enjoyable rental experience for tourists visiting Udaipur.

7.1 Assumptions

Technical Assumptions:

- All users, including visitors and registered customers, will have access to stable internet connections for smooth operation of the web portal.
- The hosting platform will ensure adequate uptime and scalability to handle peak loads during tourist seasons.
- Integration with GPS and payment gateways will work seamlessly without significant delays or compatibility issues.

User-Related Assumptions:

- Users are familiar with basic website navigation and online payment processes.
- Feedback and review sections will receive genuine and constructive inputs from users to improve services.
- Visitors will be encouraged to register for better access to features and promotions.

Operational Assumptions:

- Admins will regularly update vehicle details, pricing, and promotional offers to keep the platform relevant.
- Customer support will actively address user queries, ensuring high satisfaction rates.
- The business will continue maintaining a fleet of well-serviced and reliable vehicles.

Project Implementation Assumptions:

- The system will be deployed within a predefined timeline and budget.
- Staff will receive adequate training to manage the admin panel efficiently.
- The transition from offline to online operations will not disrupt existing services.

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7.2 Limitations

Technical Limitations:

- The system relies heavily on stable internet connectivity; disruptions can hinder user access and service efficiency.
- Limited offline capabilities might inconvenience users in areas with poor network coverage.
- The platform's initial setup may not support integration with advanced technologies like AI-based recommendations or voice commands.

User-Related Limitations:

- Some users, particularly those less tech-savvy, may find it challenging to navigate the online booking process.
- Language barriers might pose difficulties for tourists unfamiliar with English or local languages.

Operational Limitations:

- Regular maintenance and updates are required to ensure system security and reliability, which may occasionally disrupt service.
- Dependence on third-party services like payment gateways and GPS providers could lead to delays or downtime outside the company’s control.

Scalability Limitations:

- During peak tourist seasons, the system may experience slower response times or performance issues without adequate scaling infrastructure.
- The database design may require upgrades to handle large volumes of concurrent user data and transactions over time.

Feedback System Limitations:

- Automated moderation of feedback may fail to identify all spam or inappropriate reviews, affecting the credibility of the system.
- Genuine customer concerns might go unnoticed without a robust manual review process.

7.3 Future Scope

Integration with AI: Implementing AI-powered recommendation systems to suggest vehicles based on user preferences, travel history, and location.

Mobile Application: Expanding the platform with a dedicated mobile application to provide a seamless and convenient booking experience for users on the go.

Multi-Language Support: Adding support for multiple languages to cater to a diverse range of users, ensuring inclusivity and accessibility.

Dynamic Pricing: Introducing a dynamic pricing model based on demand, seasonality, and availability, ensuring competitive pricing for users.

Partnerships and Collaboration: Collaborating with local tourist agencies and hotels to offer combined travel and stay packages.

Fleet Expansion: Expanding the vehicle fleet to include luxury cars and SUVs to cater to a broader audience.

Enhanced Security Features: Incorporating biometric authentication and advanced document verification for secure bookings and transactions.

User Engagement: Developing loyalty programs and reward systems to retain customers and incentivize repeat bookings.

7.4 Conclusion

The online vehicle rental platform for Bhakti Tours and Travel is designed to make renting vehicles easier and more convenient for users. It provides a smooth booking process, secure payments, and a simple interface for both visitors and registered users. The system will also help the company manage their services better while being flexible for future updates. Overall, this project will improve customer satisfaction and help the company grow in the travel and tourism industry.

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Evaluation Form-1

Evaluation Form

Name of the student : Jainy Gandhi

Project Title : E-service platform for vehicle rentals

Name of the Project Guide : Hemant Agrawal

Designation : Owner

Organisation : Bhakti Tours and Travels

Address : 11, Shivaji Nagar, Inside Udaipole, Udaipur-313001, Rajasthan.

Contact No : 9829244434.

Kindly rate the student by ticking in the appropriate boxes.

Parameters	Needs Improvement	Average	Good	Excellent
Knowledge a) Conceptual b) Practical Application		✓		
Communication Skills				✓
Eagerness to Learn			✓	
Grasping Power			✓	
Ability to work in a team			✓	
Leadership Abilities			✓	
Punctually		✓		
Sincerity			✓	
Overall Rating of the Project Report			✓	

Remarks (If any) :-

We thank you for giving our student the opportunity to gain meaningful experience and for sharing your feedback with us.

Bhakti Tour & Travels

Signature of Project Guide & Date

Seal/Stamp of Company

Evaluation Form-2

Evaluation Form

Name of the student : Diya Jain

Project Title : E-service platform for vehicle rentals

Name of the Project Guide : Hemant Agrawal

Designation : Owner

Organisation : Bhakti Tours and Travels.

Address : 11, Shivaji Nagar, Inside Udaipole, Udaipur-313001, Rajasthan

Contact No : 9829444434

Kindly rate the student by ticking in the appropriate boxes.

Parameters	Needs Improvement	Average	Good	Excellent
Knowledge a) Conceptual b) Practical Application			✓	
Communication Skills				✓
Eagerness to Learn				✓
Grasping Power				✓
Ability to work in a team				✓
Leadership Abilities				✓
Punctually				✓
Sincerity				✓
Overall Rating of the Project Report				✓

Remarks (If any) :-

We thank you for giving our student the opportunity to gain meaningful experience and for sharing your feedback with us.

Bhakti Tour & Travels

Signature of Project Guide & Date

Seal/Stamp of Company

Evaluation Form-3

c

Evaluation Form

Name of the student : Mannan Agrawal

Project Title : E-service platform for vehicle rentals

Name of the Project Guide : Hemant Agrawal

Designation : Owner

Organisation : Bhakti Tours and Travels

Address : 11, Shivaji Nagar, Inside Udaipole, Udaipur - 313001, Rajasthan

Contact No : 9829244434

Kindly rate the student by ticking in the appropriate boxes.

Parameters	Needs Improvement	Average	Good	Excellent
Knowledge a) Conceptual b) Practical Application				✓
Communication Skills			✓	
Eagerness to Learn				✓
Grasping Power			✓	
Ability to work in a team		✓		
Leadership Abilities			✓	
Punctually		✓		
Sincerity				✓
Overall Rating of the Project Report			✓	

Remarks (If any) :-

We thank you for giving our student the opportunity to gain meaningful experience and for sharing your feedback with us.

Bhakti Tour & Travels

Signature of Project Guide & Date

Seal/Stamp of Company